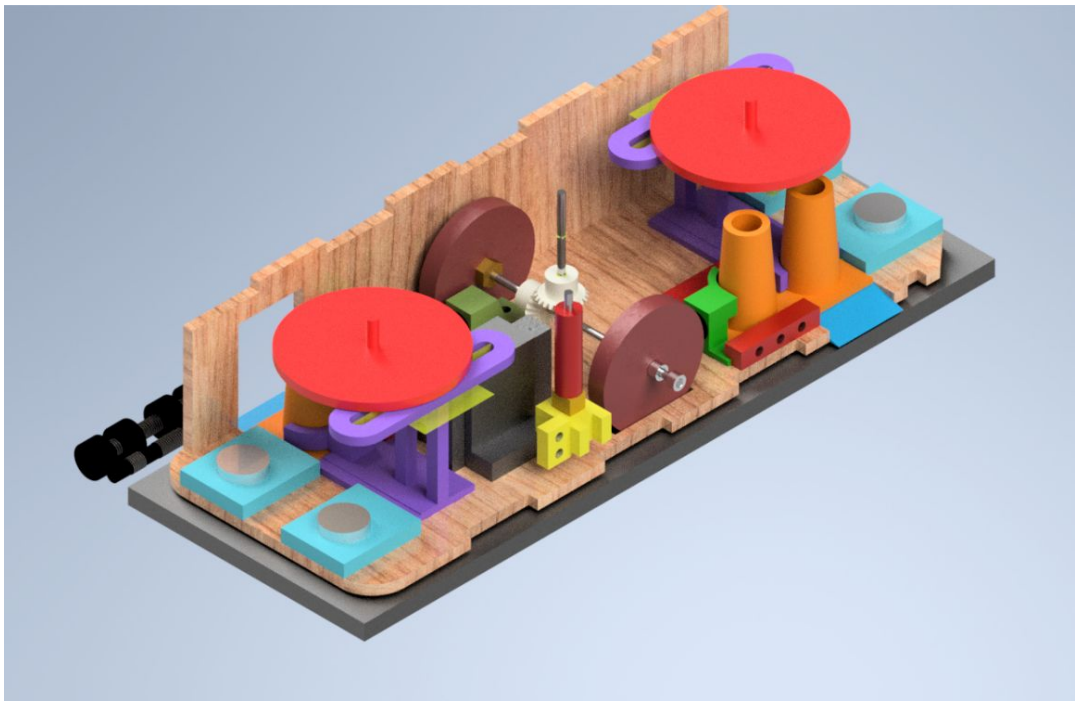
Design and Make Report

GROUP 48

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Summary

Laying traffic cones on motorways poses a dangerous task due to the high speeds vehicles that travel on motorways. This job is carried out manually and an automated alternative is yet to be fully deployed across the nation's motorways. The aim of this design project is to design an automated system that only uses mechanical elements to attain the set distances between the cones. The process started out by devising a set of requirements that the client desires.

Seven individual concept drawings were made based on the specification, which were then evaluated using the weighted score method which compared advantages and disadvantages of all the designs to choose a design that we would continue to develop. The concept that was chosen consisted of a scotch yoke mechanism which pushed the cones out of the side of the vehicle.

During the development of the prototype, numerous problems were encountered involving: the stability of the vehicle, the gear mechanism system consisting of a mutilated spur gear, determining correct gear ratios, and fitting all the machine components within the imposed specification dimensions.

Mechanical calculations were done to ensure that the appropriate gear ratios were calculated. They were essential for correctly coordinating the timings of the vehicles systems so that the design would function as designed. Finding the correct gear ratios for the mechanism allow accurate placement of the cones. The motor was then chosen based on the desired completion time and the torque values.

Cost minimizing was key during this project. The parts purchased from approved external suppliers came up to a total of £71.61 and the total cost of items from the university supply is £2.84. This means the total resource price was £74.45. Labour costs came to a total of £835. The total cost of all the different expenditures was £1027.77, but much of this came from labor costs.

The prototype was within the £90 external supplier resource budget and the theoretical requirements in the product design specification set for the vehicle by the client were met.

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1 – Introduction

If a lane on a motorway must be closed, road workers must lay cones to make the lane unavailable to oncoming traffic. These types of operations are usually carried out while motorways are open, meaning that contraflow lanes must be created with these cones. In the past this has been done manually, a dangerous task for workers with high-speed vehicles passing by. To nullify this danger, automated cone dispensers can be used. One of the first automated cone dispensers was advertised in Switzerland in 2013 (Verdegro Group, 2013). Now, automated cone laying machines are widely used, such as the machine from Highways Industry in England in 2020 as shown in Figure 1. The objective of this project is to design a self-propelled vehicle that stores and dispenses traffic cones on a smaller scale, by exploring mechanical solutions for the complex process of unloading cones.



Figure 1 Highways industry automatic cone laying vehicle [1]

1.2 - Design Brief

The vehicle must be able to place cones on a road at ground level in the double line arrangement shown in in Figure 2, where the vehicle moves from the right of figure 1 and travel to the left-hand side and travel to point B. The dimensions of the cone positions relative to each other are shown. The process should take no longer than a minute to lay out 4 cones. Whilst only 4 cones are required to be placed onto the road in this project, a design which allows for continued cone placement is ideal. The blue dashed and dotted line is the centre of the road being formed.

All distances are to and from the centre of the cones. The vehicle is to be powered by a small electric motor and a single battery (two motors for a stacked cone system). The motor in question cannot be subject to any axial or radial forces. The chassis dimensions can be at a maximum of 380mm height, 180mm width and 300mm height.

Following the design, a working prototype will be constructed. Parts may be externally sourced or made bespoke. When designing and constructing the prototype, mechanical fasteners should be used as opposed to glue.

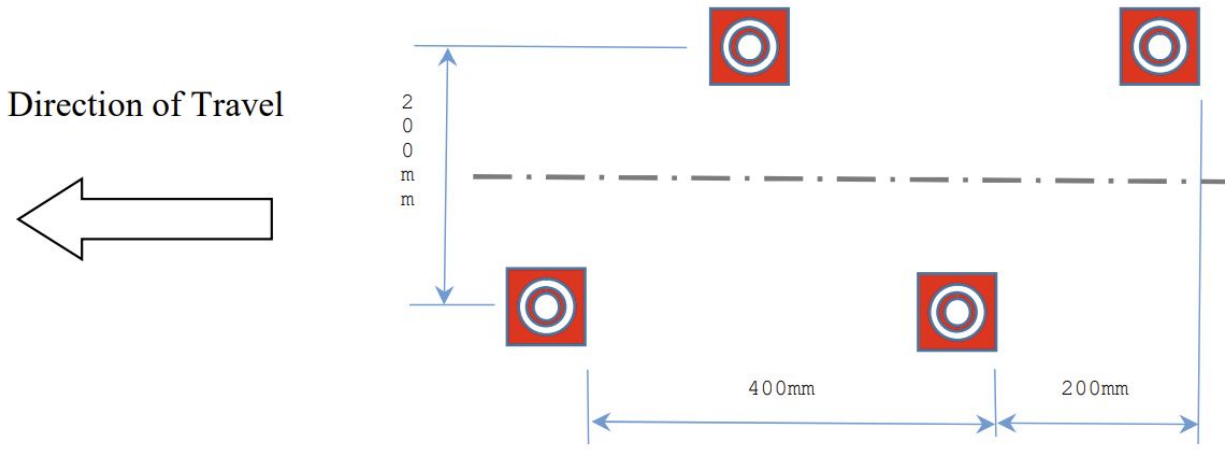


Figure 2 shows the desired double line cone placement pattern.

1.3 – Design Process

Figure 3 conveys each stage of the design process. The first step is to understand the design brief and the relevant specification requirements as shown in the next section of this report, the product design specification. The information extracted is used to set the key requirements for the prototype. Each member of the team then devises a concept and produces an isometric concept drawing. This is followed by the concept selection stage, where a concept is selected using one of multiple selection methods (e.g. controlled convergence and multi-criteria decision analysis). The embodiment design phase consists of making justified calculations for each subsystem involved in the selected concept. The design can now be modelled on CAD. Details such as joining components and assembly plans are also explored. The final step is to purchase the necessary equipment and calculate the total costs such as labour, equipment, and assembly costs.

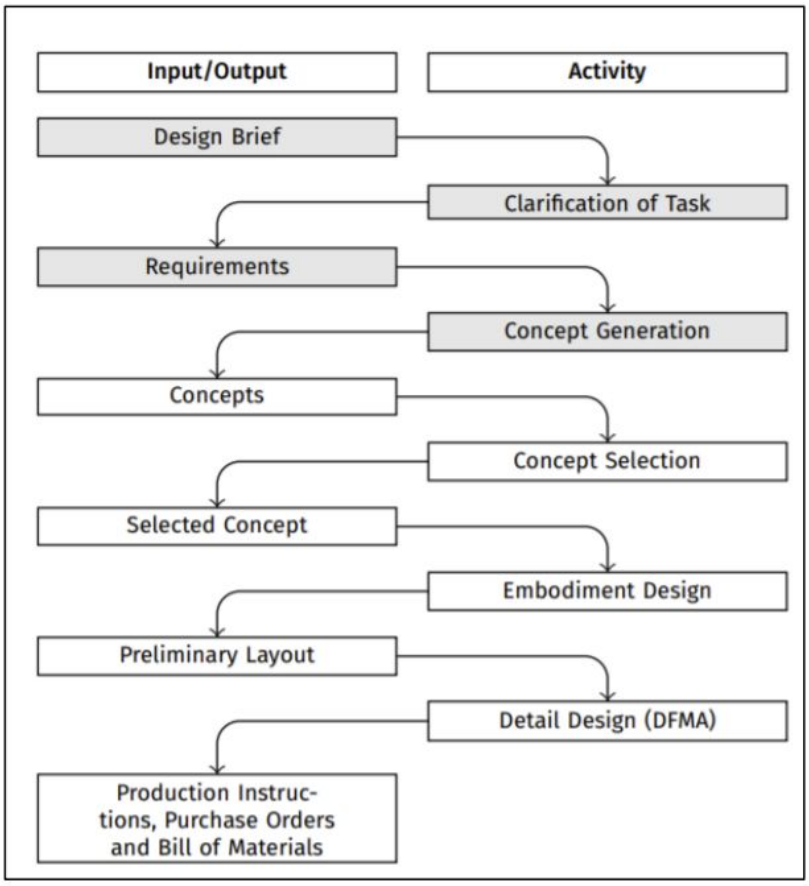


Figure 3 shows the Flow Chart Depicting the Design Process [2]

2 - Product Design Specification

D/W	Wt	Requirement	Specification for Cone Unloader	Date 20/03/23 Modified:	Source
		Function			
D			Cones unloaded to ground level.		Design Brief
D			Cones unloaded upright.		Design Brief
D			Lays 4 cones in no more than 60 seconds.		R.V Q&A
D			Cones unloaded in double line pattern (see Figure 2).		Design Brief
D			Minimum of 4 cones to be unloaded.		R.V Q&A
W	H		Extensible prototype that can carry and unload more than 4 cones.		R.V Q&A
W	H		Reliable.		Design Brief
		Design Specification			
D			Maximum 180 mm width.		Design Brief
D			Maximum 380 mm length.		Design Brief
D			Maximum 300 mm height.		Design Brief
D			Battery powered and self-propelled.		R.V Q&A
D			No radial nor axial loads subjected to motor		Design Brief
D			Powered by no more than 2 DC electric motors		Design Brief
D			Motors of maximum voltage 9V to be used.		R.V Q&A
D			No glue/adhesives used.		Design Brief

D			Only mechanical fasteners are used.		Design Brief
D			Controlled only by mechanical elements.		Design Brief
			All shafts are 4 mm diameter.	27/03	R.V.
		Economics			
D			£90 limit on externally ordered parts.		Q&A
D			Only buy parts from approved suppliers or made in house.		Q&A

Table 1 shows the product design specification for the cone unloader.

3 - Concept Drawings

Using the design specification, seven concept drawings were created and presented using isometric sketches. Each design used a different method of placing the cones and a chosen mechanism to power the process. Below, each concept is explained in terms of its workings and how the design met some of the specification criteria.

3.1 - Concept 1: Luca Piergentili

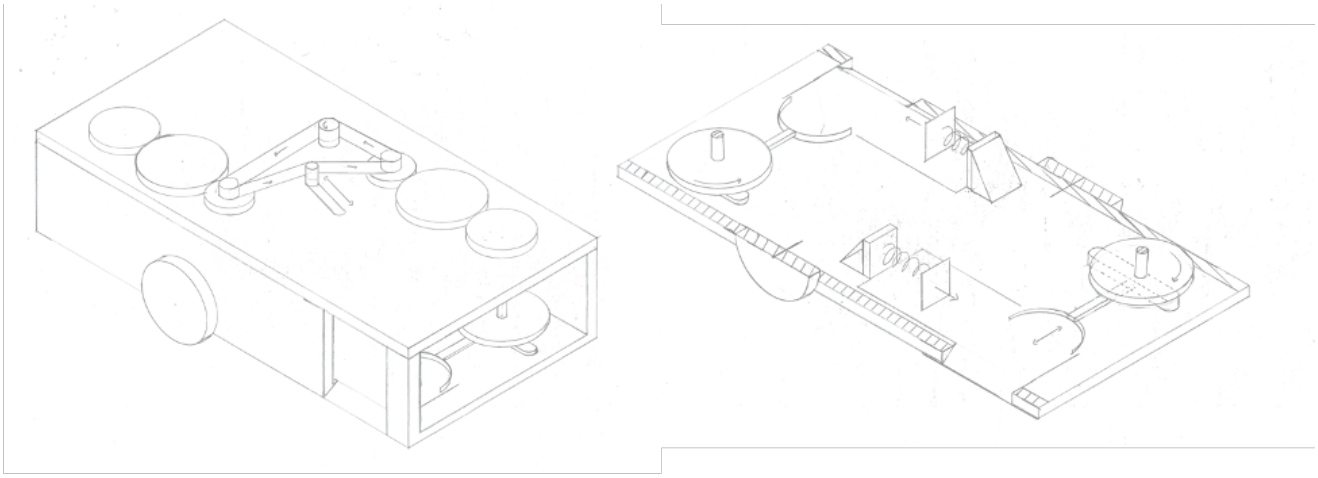


Figure 4 showing Luca's isometric concept (concept 1)

The concept seen in Figure 4 was drawn by Luca but was initially conceptualized by Elie, James, and Luca during the first meeting for this project. The design was later developed by Luca and became what is seen above. It consists of two levels, a mechanism level below as seen on the right, and a gear level on the top as seen on the left. The cones are pushed out at either side of the vehicle by a scotch yoke mechanism which is connected via a shaft to the gears on the upper level.

The two exit points of the cones are distanced so that the cones are placed in the correct spots to create the parallelogram pattern, this means that the vehicle only has to stop and perform the cone ejection twice in order to complete the full cone pattern.

Other cones are stored in the machine in magazine-like contraptions with spring pushers pushing them towards the ejection channel. Once one cone is ejected, the next is pushed into place ready to be ejected. This is displayed in Figure 5 which was done on an iPad using an e-notebook software. The plunger and the spring pusher have shapes made so that they can interlock so that they do not interfere with each other's movement.

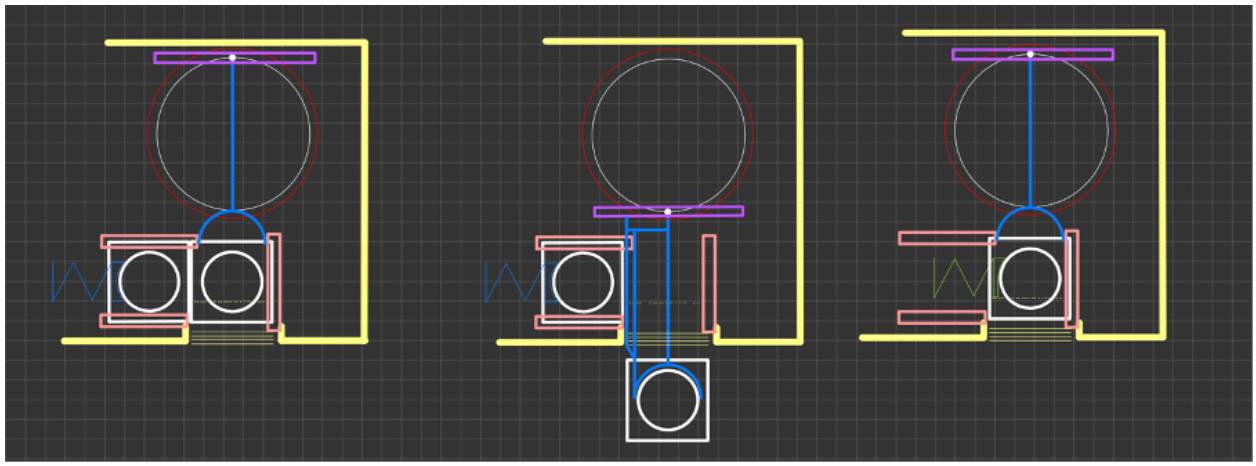


Figure 5 showing the cone ejection sequence for concept 1

The top level houses the gear mechanism. It is composed of a driven gear, that is connected to a motor via a shaft below it, that drives the whole mechanism using a belt. The belt is connected to the gear connected to the wheels (the "wheel gear"), which is connected to a mutilated gear with half the teeth removed, which connects to the gear turning the scotch yoke (the "scotch yoke" gear). The belt would be kept in tension using a sliding idler so, this would allow some flexibility when choosing belt lengths as with this system more sizes would be able to be accommodated and it would be easier to remove and place the belt.

3.2 - Concept 2: Amith Kutti Bala

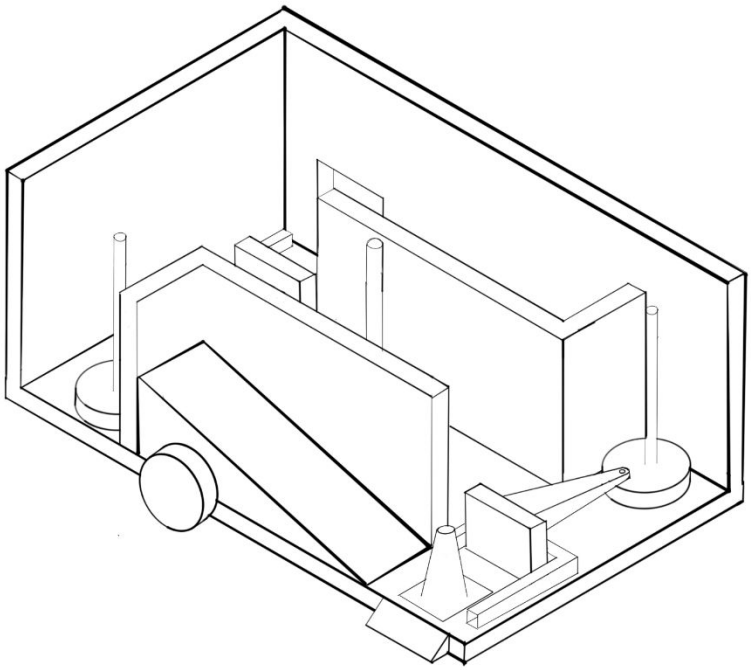


Figure 6.a showing Amith's isometric concept (Standard view)

The design concept shown in figure 6 consists 0f 2 levels; a base level c0nsisting of the main mechanisms and a second level consisting of the gears. The design uses gravity to it's advantage by loading cones on a sloped surface. The cones thus slide down to the bottom of the ramp and stack themselves horizontally. As the vehicle stops at the appropriate location, the comes are dispensed out through the help of a crank mechanism which ensures that the comes are placed outside the vehicle. In order to reduce the stoppage time to place each cone, the chassis is modified so that the c0nes are dispensed from two locations, and the distance between the both are according to the distance of separation mentioned in the pattern. This, the vehicle will only have to stop twice to undergo the task. Furthermore, extra supports and railings are provided in necessary locations to ensure smooth running in the mechanism. The second or top level contains the gear mechanism which is connected to the crank and the wheels by vertical shafts from the main level. Figure 6.b shows a deconstructed view of the main components and mechanism system, while figure 6.c givved a simple representation of what the gear mechanism could look like. It consists of two crank gears connected to a series of gears for ensuring the right

ratio, which is also connected to a motor to power the vehicle's wheels on the outside of the chassis.

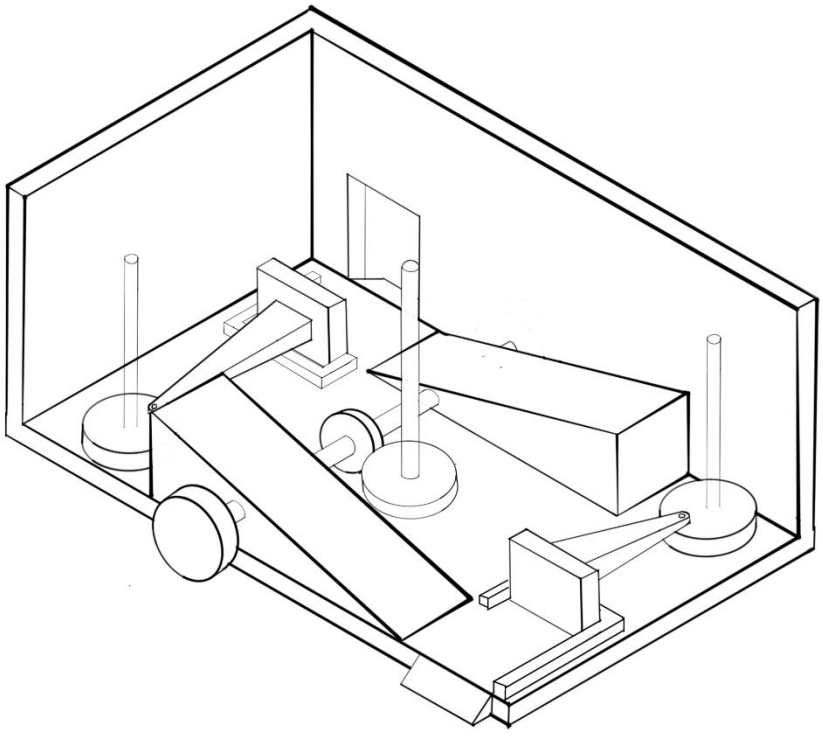


Figure 6.b showing Amith's isometric concept (Deconstructed view)

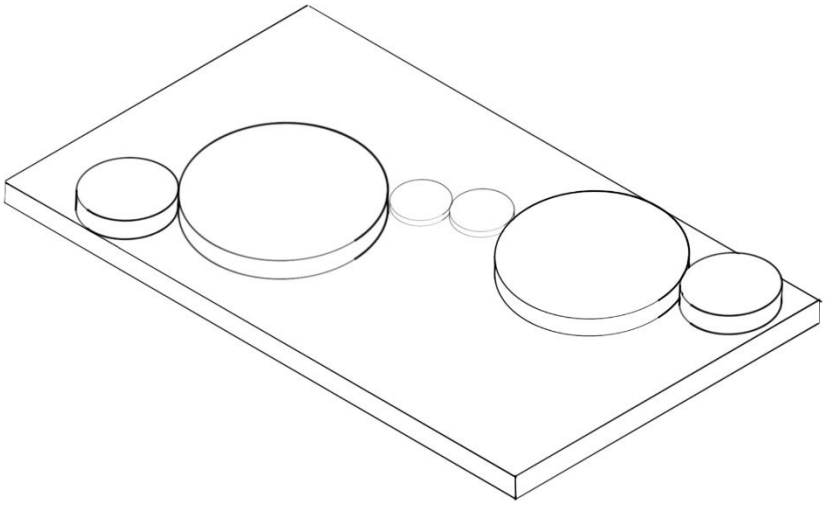


Figure 6.c showing Amith's isometric concept (second level)

3.3 - Concept 3: Israt Jahan Pia

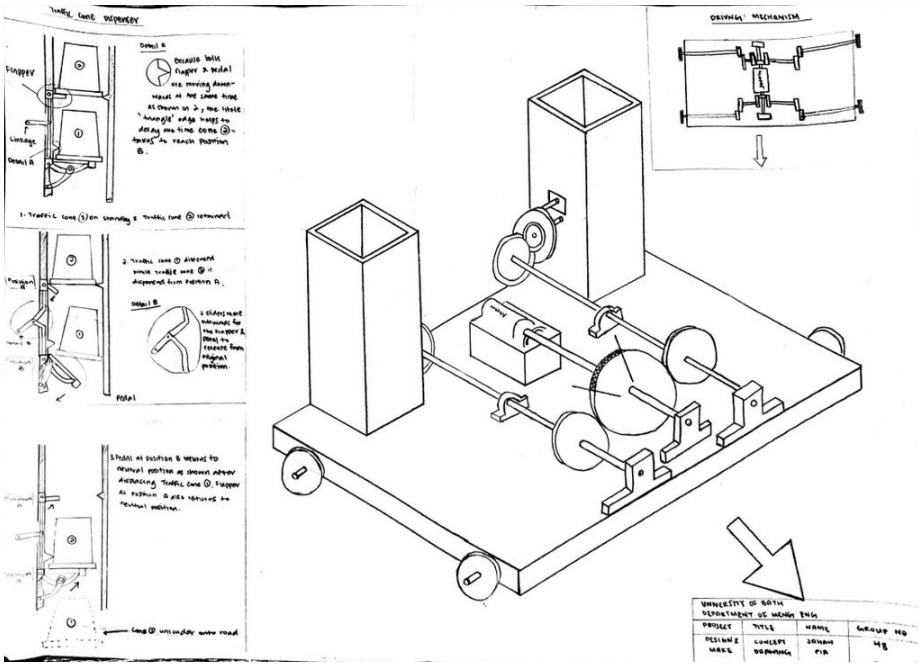


Figure 7 showing Pia's isometric concept (concept 1)

This design was inspired by a vending machine. As shown in the Figure, two columns were made at both sides as the cones must be displaced at both sides of the truck at 200m apart from each other. In this design, each column holds two cones. To allow the cones to drop at its respective places accurately, the columns were designed with a cut-out at the bottom with a height of 55mm. This also ensured that the individual cones would not collide with the wall of column when being dropped to the ground.

Each column consists of a flapper and a pedal to ensure a smooth delivery of displacing cones accurately. The columns are designed with a flapper and pedal, which are connected by a linking mechanism. On both sides of columns, 'triangles' were created to prolong the time a cone takes to reach Position B from Position A. The flapper and pedal are designed to move at the same time hence the 'triangles' would prevent cone 2 to reach Position B when the pedal is retracted.

3.4 - Concept 4: James Carter

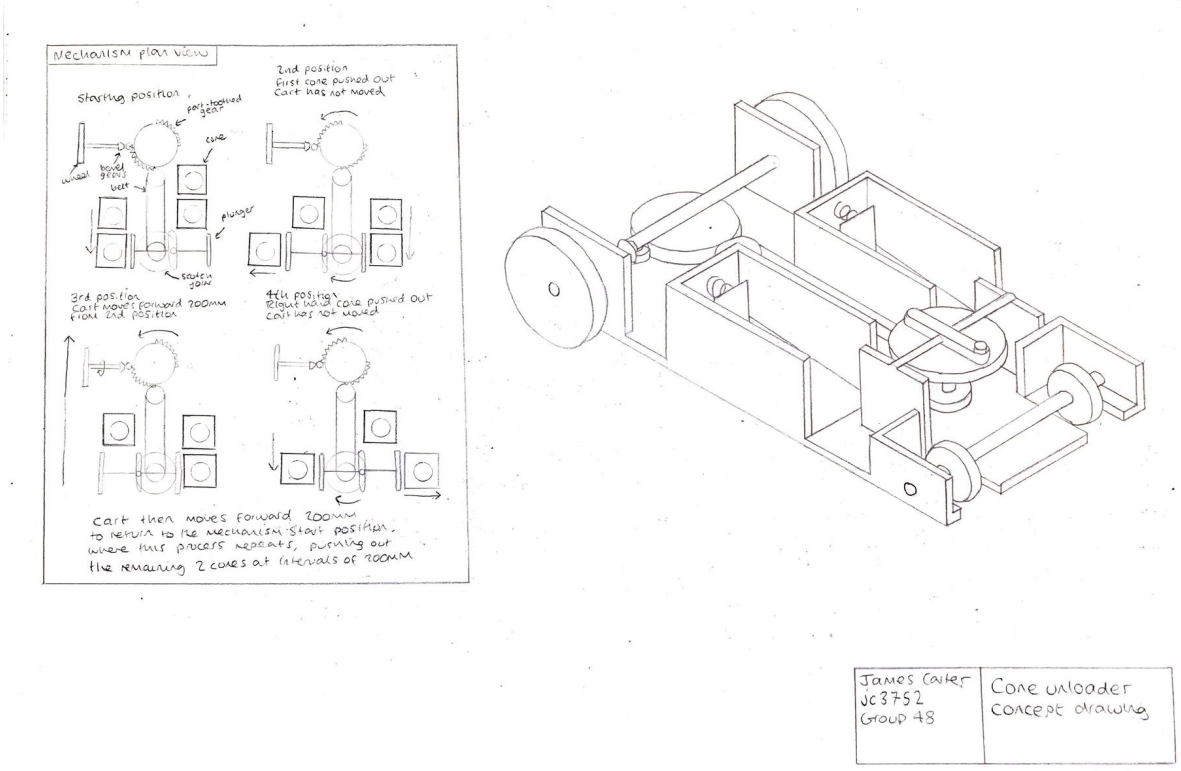


Figure 8 showing James's isometric concept (concept 2)

This concept features a single scotch yoke mechanism which is responsible for pushing cones out on both sides intermittently. The mechanism consists of a mutilated gear connected to wheels through bevel gears and a scotch yoke through a long belt. This mechanism, as shown in the mechanism plan view, results in a process where the scotch yoke completes half a rotation, thereby pushing out a cone on one side. The mutilated gear then disengages with the scotch yoke and engages with the wheels, which moves the vehicle forward 200mm. The wheels are then disengaged whilst the scotch yoke once again becomes active, performing half a rotation, thereby pushing the cone out on the other side. The cones are loaded into their ejection bays by a spring once the ejection bay becomes cleared. The intermittent nature of the cones only being pushed out when the vehicle is stationary means the cones are more likely to be positioned correctly. With this concept, only 1 motor would be needed. It is not shown in this concept drawing, however it would be connected to the large, mutilated gear. The difficulty with this design would be fitting all the parts, specifically the scotch yoke and plungers, into the allotted maximum dimensions for this project. Whilst the scotch yoke was used in other concepts.

3.5 - Concept 5: Elie Barret

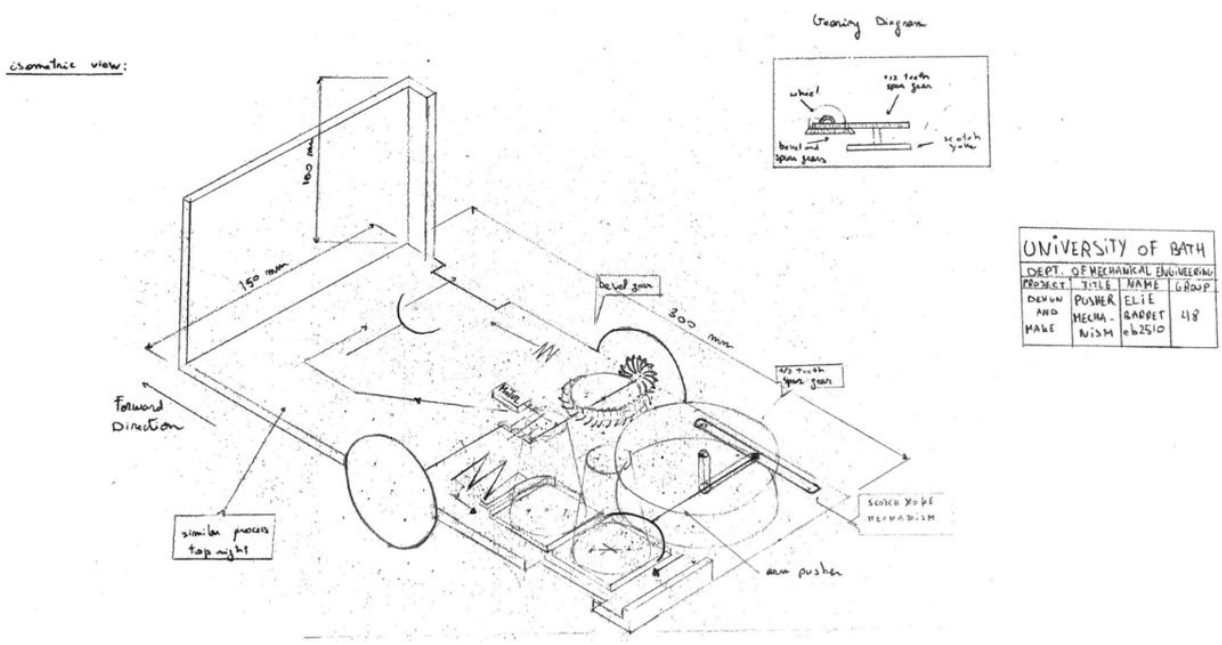


Figure 9 showing Elie's isometric concept (concept 2)

This concept consists of two levels: a mechanism level below, and a gear level on the top as seen on Figure 9. The cones are pushed out at either side of the vehicle by a scotch yoke mechanism which is connected via a shaft to the gears on the upper level. The spur gear on the upper level is a half teeth spur gear that allows the rest of the mechanism to stop to push the four cones out through the two exits. Moreover, the latter makes turn another spur gear (this time full teeth) connected to a bevel gear that is transforming the motion from horizontal to vertical by being connected to another bevel gear connected to the wheels of the machine by a shaft. An electric motor linked to the half teeth spur gear is powering the mechanism.

There are two exit points of the cones, on the bottom left and top right of the machine. They are distanced so that the cones are placed in the correct spots to create the parallelogram cone pattern by making the vehicle only stop and perform the cone ejection twice. Once one cone is ejected, the next is pushed by a spring into place ready to be ejected.

(The concept seen in Figure 8 was drawn by Elie but was initially conceptualized by Luca, James, and Elie during the first meeting for this project. The design was later developed by Elie and became what is seen above.)

3.6 - Concept 6: Advik Agarwal

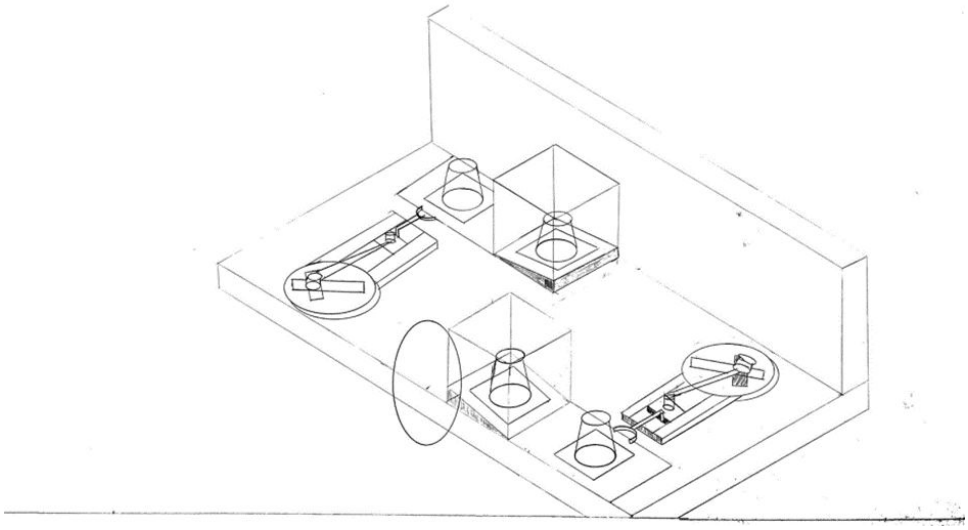


Figure 10 shows Advik's isometric concept (concept 6)

This design features a supporting side rail that extends from the mover to prevent it from toppling over when a cone is ejected from the placer. The side rail helps maintain the stability of the mover and ensures that it remains upright during the ejection process.

The slider crank mechanism used in the design converts rotary motion into linear motion, producing the force needed to eject the cone from the placer. This mechanism is augmented with a Whitworth quick return mechanism that ensures rapid retraction of the plunger after ejection, preventing interference with the other cone. The platform beneath the cone is inclined at a 35-degree angle and covered with a plastic film to reduce friction, enabling the cone to slide smoothly down. The mover is powered by a motor at its centre that drives two large wheels on either side via various gears. The speed of the mover can be adjusted as needed to optimize its performance.

To prevent toppling, smaller supporting wheels are placed at the front and back of the middle. The supporting side rail extends from the mover to provide additional stability during the ejection process.

3.7 - Concept 7: Yigit Ersahin

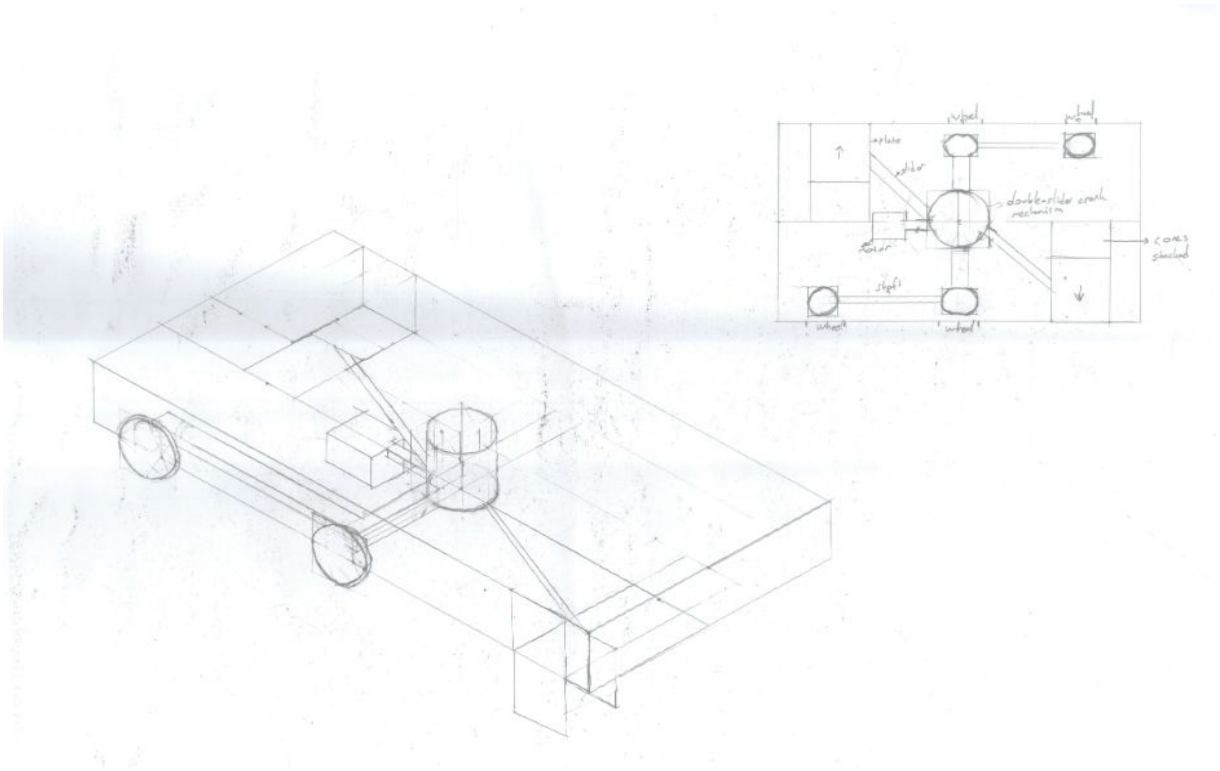


Figure 11 shows Yigit's isometric concept (concept 7)

In this concept, a double-slider crank mechanism is used. Two connecting rods and two sliders are connected to the crank in a double-slider crank mechanism. One slider is upon the other. Cones are stacked at the two opposite ends of the mechanism. Sliders are connected to plates. The mechanism works as slider crank mechanisms slide the plates under cones, and as the bottom cone starts to drop, the plate connected to the slider comes back and blocks the way to the upper cone. Semi-tooth gears are used in this mechanism for timing, This is inspired by Luca's concept. Semi-tooth gear is attached to the motor. Semi-tooth gear is connected to the slider crank mechanism and the shaft. The shaft is connected to the wheel gears. When the cones start to drop, the vehicle stops. After the cones are dropped, the vehicle starts moving.

4 - Concept Evaluation and Selection

Following the production of the six concept drawings labelled in Table 2, each concept was compared in distinct aspects such as continuousness and complexity, using weighted scores as shown in Figure .

	Name	Description
Concept 1	Luca	2 Scotch yokes, Toothed belt
Concept 2	Amith	Double Slider crank
Concept 3	Pia	Column cone dropper
Concept 4	James	Scotch yoke with aligned cones
Concept 5	Elie	Scotch yoke, bevel gears
Concept 6	Advik	Slider crank, Whitworth mechanism
Concept 7	Yigit	Double slider crank

Table 2 Description of different concepts

Out of the 7 concepts presented in section 3, Concept 1, the design was chosen due to a variety of reasons including, but not limited to, the score assigned using the weighted score method and the ability to develop the design and address and resolve any issues that arose. There were a large variety of factors that were considered when assessing the concepts seen in Section 3. All factors, along with each concept's respective scores against these factors, can be found above:

	Weighting	Concept 1	Concept 2	Concept 3	Concept 4	Concept 5	Concept 6	Concept 7
Cost	2	-4	-2	0	-2	-4	0	2
Reliability	3	6	0	-3	3	0	-3	-6
Complexity	2	-4	0	-6	-2	-4	0	2
Continuous	3	6	3	9	3	6	3	-3
Efficiency	2	6	2	4	2	4	2	2
Size	2	0	2	1	0	4	4	4
Durability	1	3	2	0	3	3	2	2
Ease of Use	2	2	0	0	2	-2	0	2
Speed	1	3	2	1	1	3	1	1
Ease of Manufacture	2	-2	0	0	0	-2	0	4
Cone Placing Accuracy	3	9	6	9	6	6	3	3
	TOTAL	25	15	15	16	14	12	13

Table 3 Weighted Evaluation Table

The weighted score method, from Clark-Carter [3] involves multiplying each data point in a set by a value, determined by the importance of that point. Sets with a greater level of importance were

given a greater weighting value, for example the cone placing accuracy, was given a higher weighted score than the durability.
Concepts were given a rating from -3 to 3 in each variable, with this then multiplied by the weight, which was scored from 1 to 3. The weighted score method was used so that the concept design flaws could be easily identified and rectified. In addition, by using the weighted score, design choices that had a higher weighted score were incorporated into the final design.

The method of weightings was beneficial as it was used to select the concept that was developed further. As shown in Table 2, Concept 1 had the greatest total score due to excelling in cone placing accuracy (the cones would be directed to the desired positions via the slope), ease of use (the cones would be stored in an individual location, to reduce the need of a worker) and continuousness; these were the areas given a higher weighting.

When comparing Concepts 1 and 5, they have multiple similarities in design and mechanical processes, because both concepts are a personal development of an initial concept idea worked by Luca, James and Elie during the first meeting. Concept 1 was chosen to be developed further due to being superior in multiple categories than Concept 5 such as Reliability, cone placing accuracy, ease of use and efficiency.

5 - Design Development

As design 1 was chosen to be the design that we would move forward with, it was important to determine and mitigate potential flaws, find possible ways to minimise the complexity of the system, and ensure the mechanism allows for a successful cone dropping process.

5.1- Mechanism correction

In concept 1, the belt shown to be attached to the wheel gears. In practice, this would cause the mechanism to permanently halt as soon as the mutilated gears disengaged with the driven wheel

gears. To solve this, the belt should be attached to the mutilated gears, so that the mechanism will be continuous. The amended mechanism layout is shown in the 2D drawing on Figure 12.

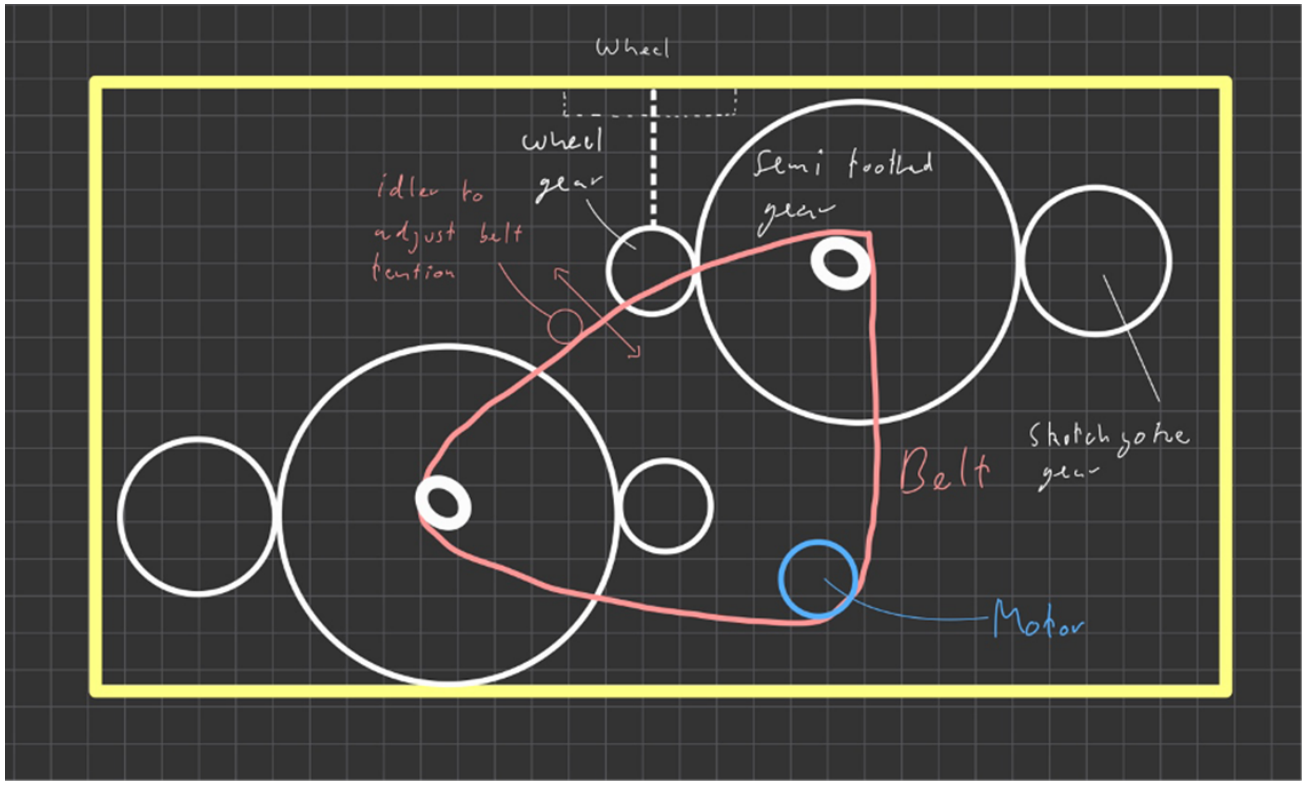


Figure 12 showing the gear mechanism on the top of the housing for concept 1.

5.2 – Mechanism timing and gear ratio calculation

One-half of the total gear mechanism is shown in Figure 13, a sketch of the mechanism was used to visually determine the gear ratios, to perfect the timing of the operations required of the vehicle. By making the mutilated gear semi-toothed, it would mean that only one of the wheel gears and the scotch yoke gear would ever be engaged. This is important as it meant that whilst the cones were ejected by the scotch yoke, the vehicle could never be in transit. It was desirable for this to be the case to ensure cone placement precision, as if the vehicle was moving, the cone could potentially catch on the edges of the vehicle, leading to either misplacement or rotation of the cones.

The gear ratio from the mutilated gear to the scotch yoke gear was 2:1, since the scotch yoke would need to complete one full rotation to move the plunger to its full extension and back to its original position during the time that the teethed half of the mutilated gear was engaged with the scotch yoke.

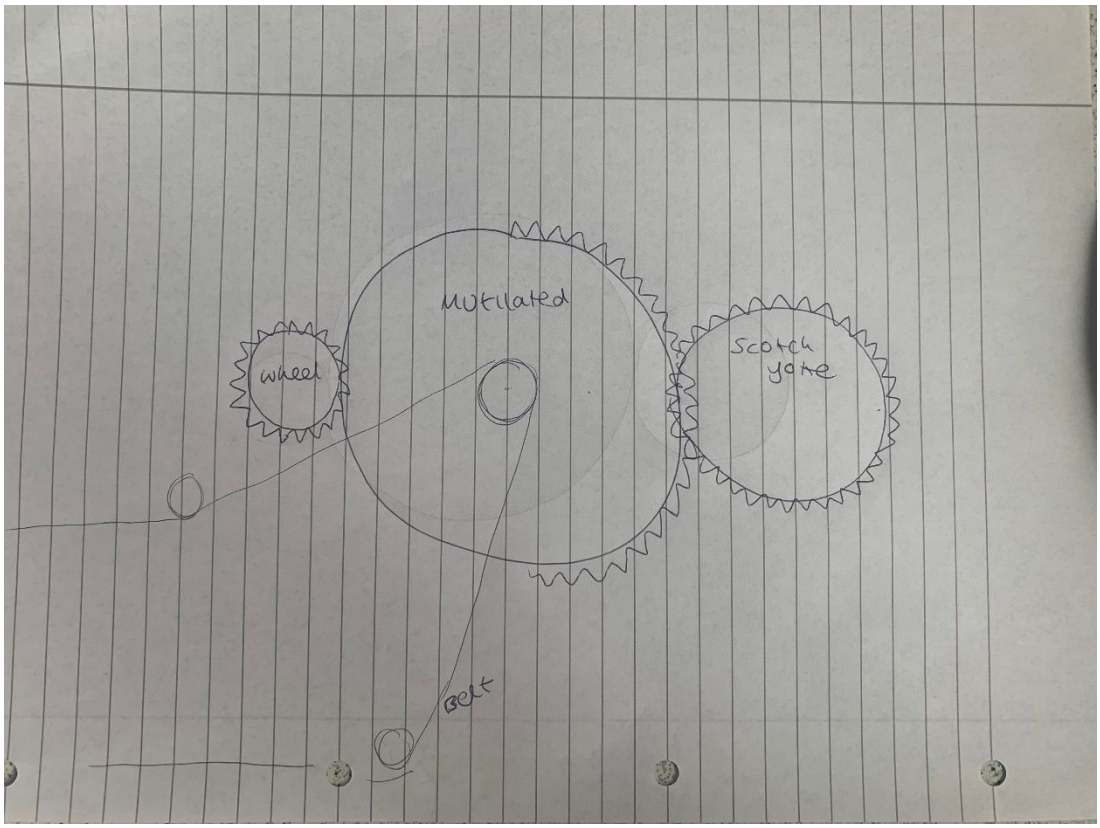


Figure 13 showing a sketch of the gear mechanism on the top of the housing for the chosen design

The gear ratio between the mutilated gear and the wheel gear was 4:1, which would mean that whilst the teeth were engaged with the wheel gears, the wheels would fully rotate twice. From this, the wheel sizes could be calculated. The vehicle was required to move 400mm between cone ejections, so to enable this, the wheel circumference would have to be 200mm, leading to a calculated wheel diameter of 63.66 mm.

Whilst the wheel size is important, the size of the gears are not, only their comparative ratios are. The sizes of gears chosen were determined by how much space was available for them between the cone ejection point and the wheels. But as the design progressed, we realised that only one

wheel gear was required as the two wheels would now be in line with each other as the chassis was made longer in order to drop the cones at a correct length from each other. This meant that one of the two mechanisms would be connected to the wheel gear and one would not. But as the design was being reproduced in Inventor, the placement of the Scotch Yoke Mechanism was finalized and its placement would have led the two gear mechanisms to clash. To fix this, one of the gear mechanisms was raised above the other so that they overlapped and did not come into contact with each other. This is shown in Figure 14.

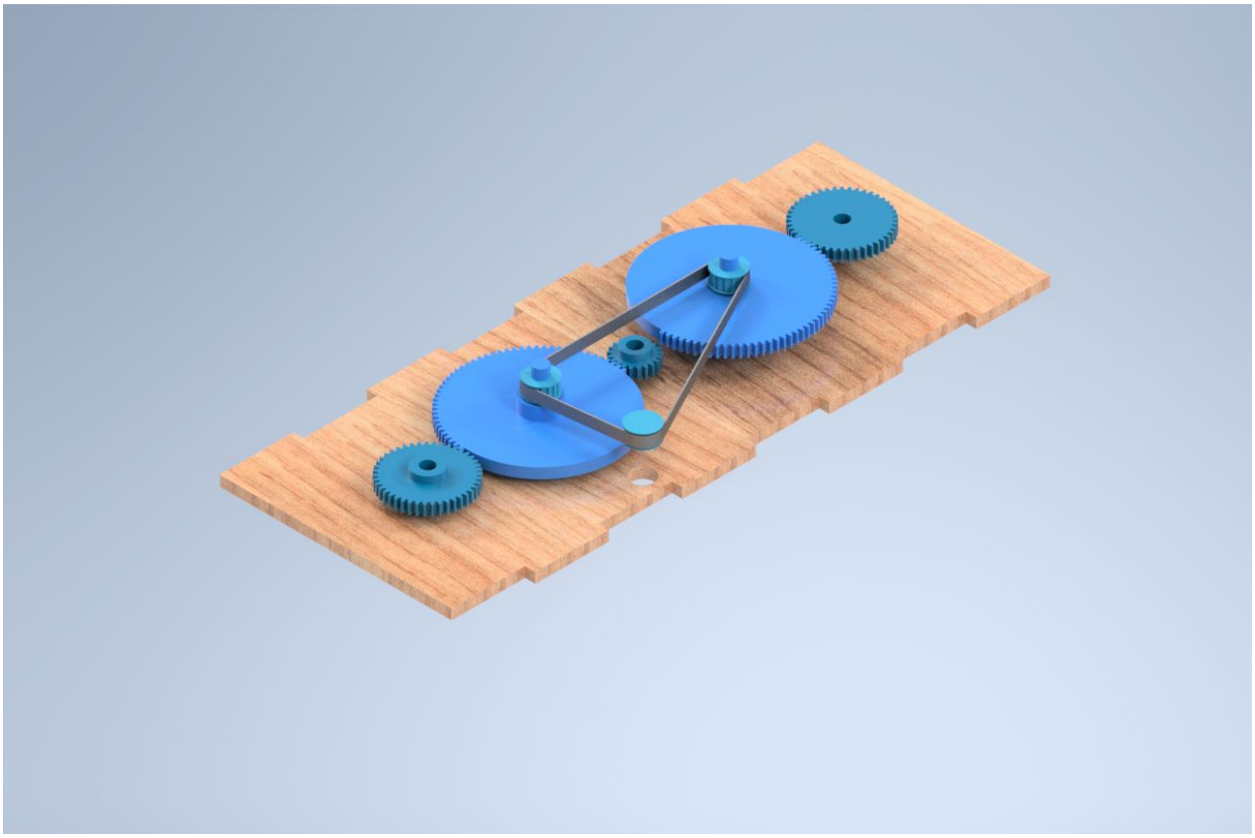


Figure 14. Gear mechanism on top of the chassis

The two connected identical mechanisms at either end of the vehicle are connected using a timed belt. It is vital that this belt is timed otherwise the processes of the cone ejection and the wheels turning will not be aligned, which will cause issues such as the vehicle to turn and cones to be positioned at incorrect locations.

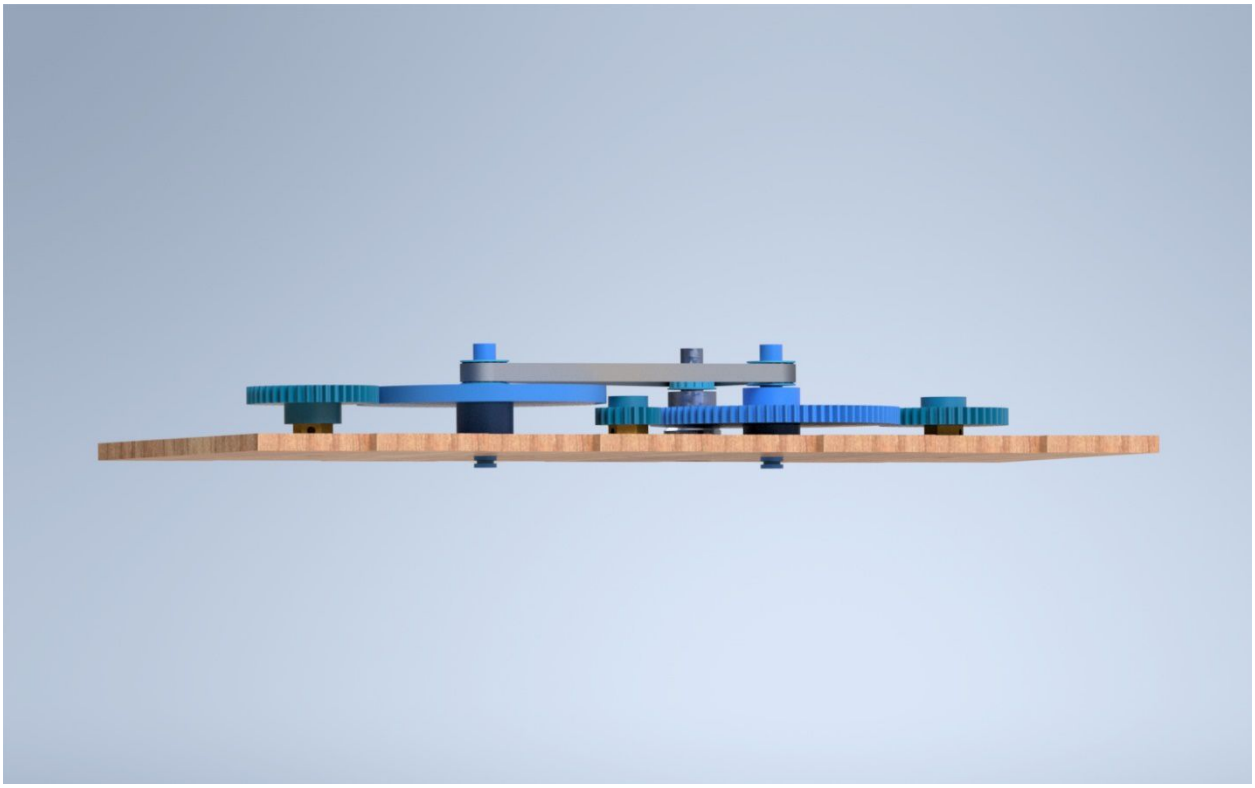


Figure 15. Gear mechanism on top of the chassis (Side View)

5.3 – Preventing cone toppling

Holding cones straight was another challenge to achieve. The cones are collocated. An uncontrolled push coming from the spring pusher or the plunger to cones would make cones tip over. To prevent this, rails are laid upon the cones. The plunger is designed in a semi-circular shape that fits around the cone which helps to distribute the force across the cone more equally and holds the cones straight and it also has a flat base at the bottom in order to push the cones from their base as well. Ramps are designed at the ejection points of the cones to prevent the cones from toppling as they are ejected from the vehicle. When the centre of mass of the cones cross an elevated edge, the cone will fall over, whilst with a ramp, the cones will slide down without the risk of toppling. The ramp is to be attached to the chassis using pins that will be interference fitted into the chassis and the ramp.

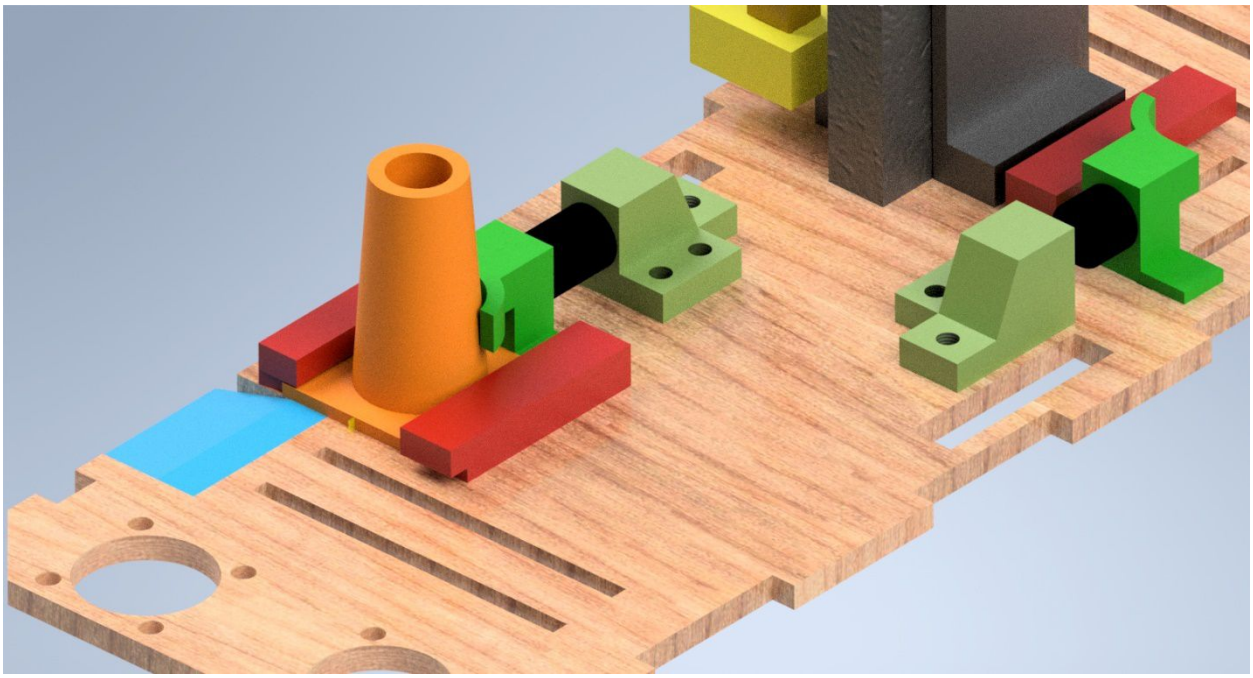


Figure 16. Rails installed to prevent cone toppling

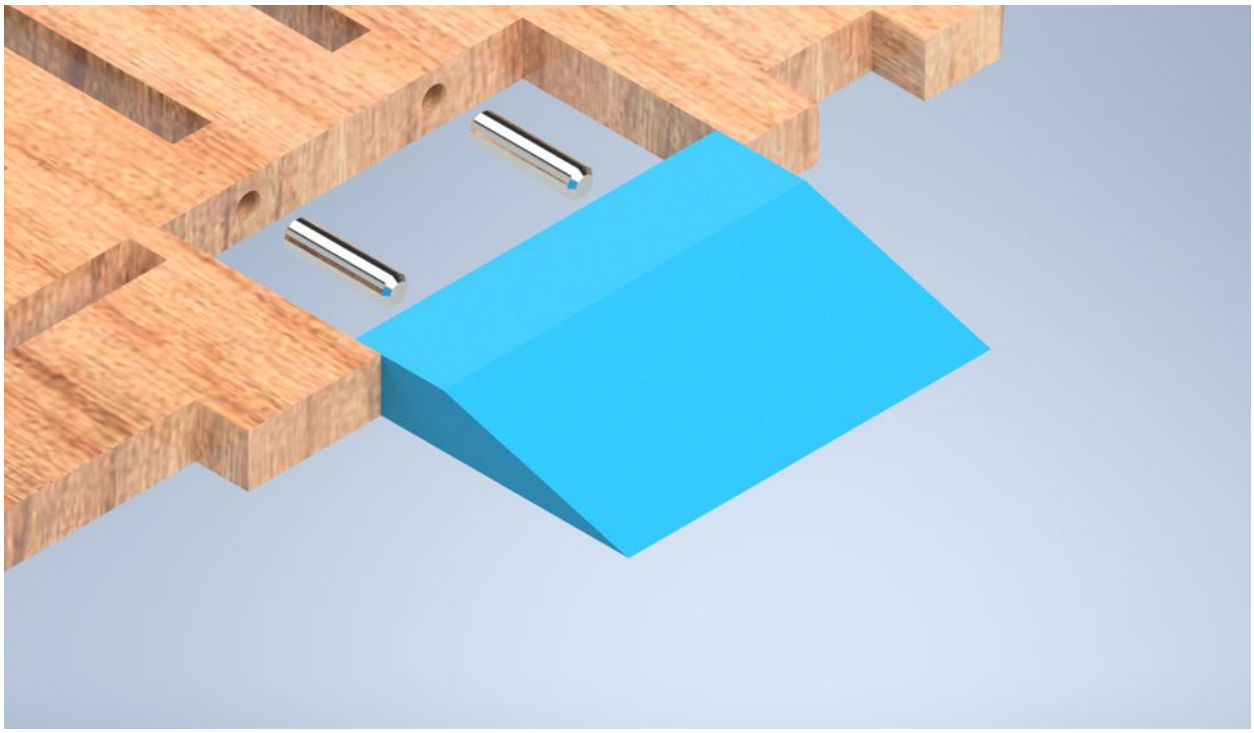


Figure 17. Ramps to push the cone smoothly into the ground

5.4 – Cone replacement mechanism

The figures below show the three phases of the ejection mechanism. The plunger was designed with an extended flat base to prevent the second cone from moving into the ejection channel whilst the first cone was being extended. Once retracted, the plunger would allow the cone to be pushed into position by the pusher. The plunger and pusher were also designed to interlock so as to not hinder each other's functionality.

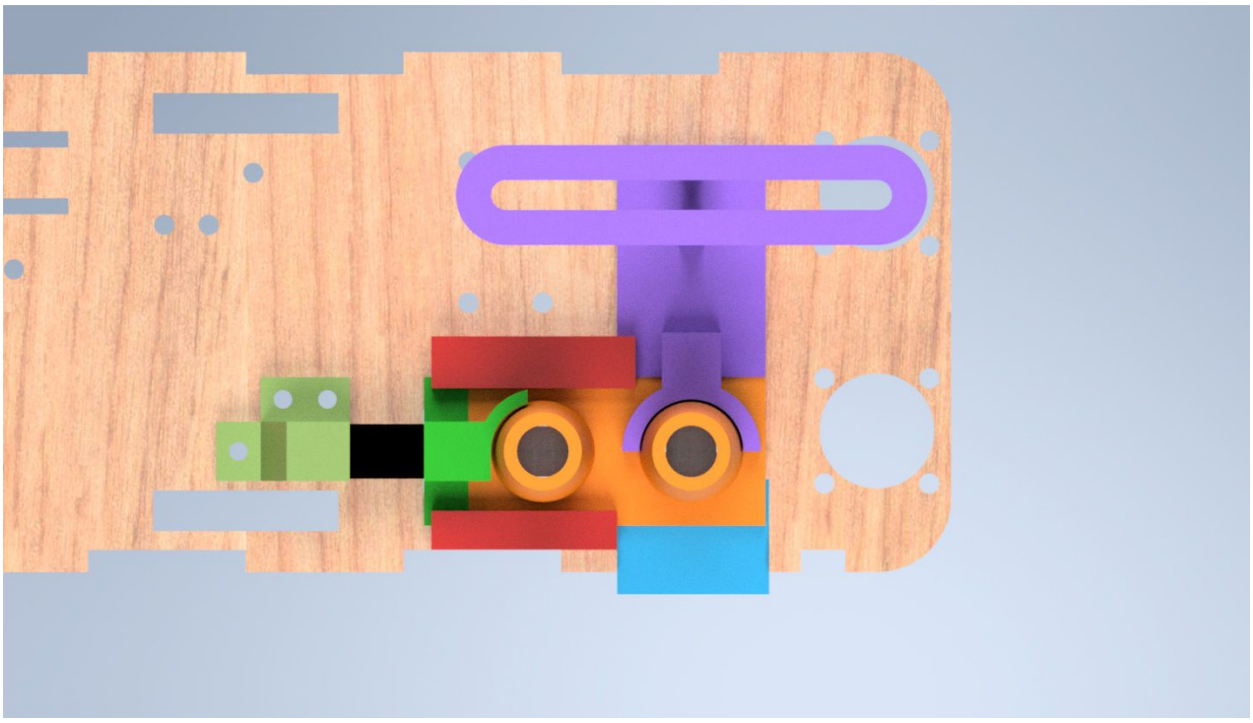


Figure 17. Plan View of the mechanism

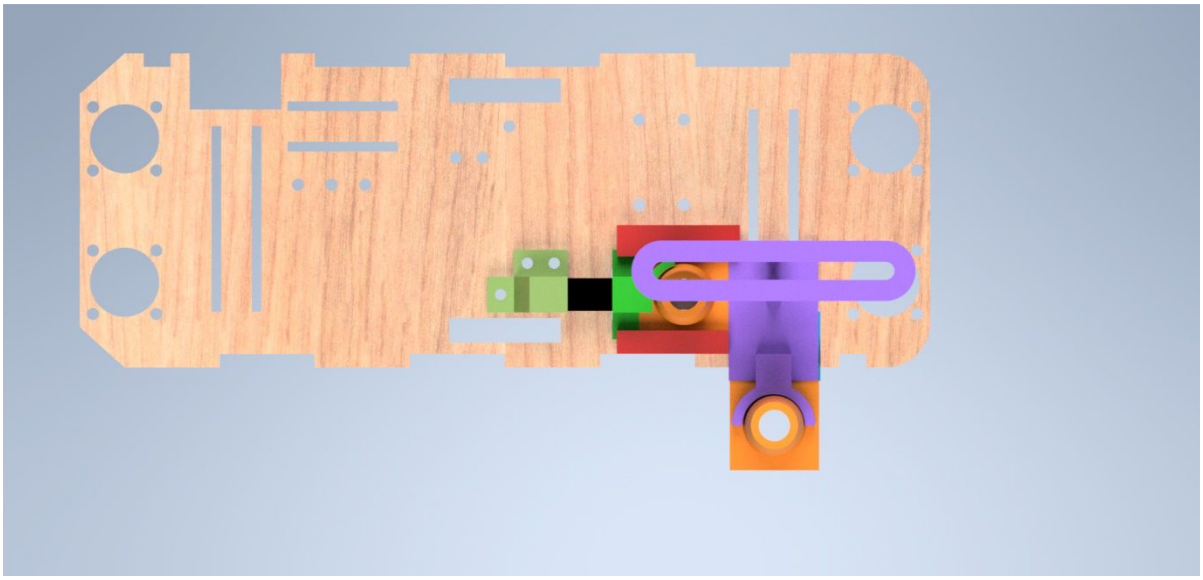


Figure 18. The cone being pushed out by the plunger

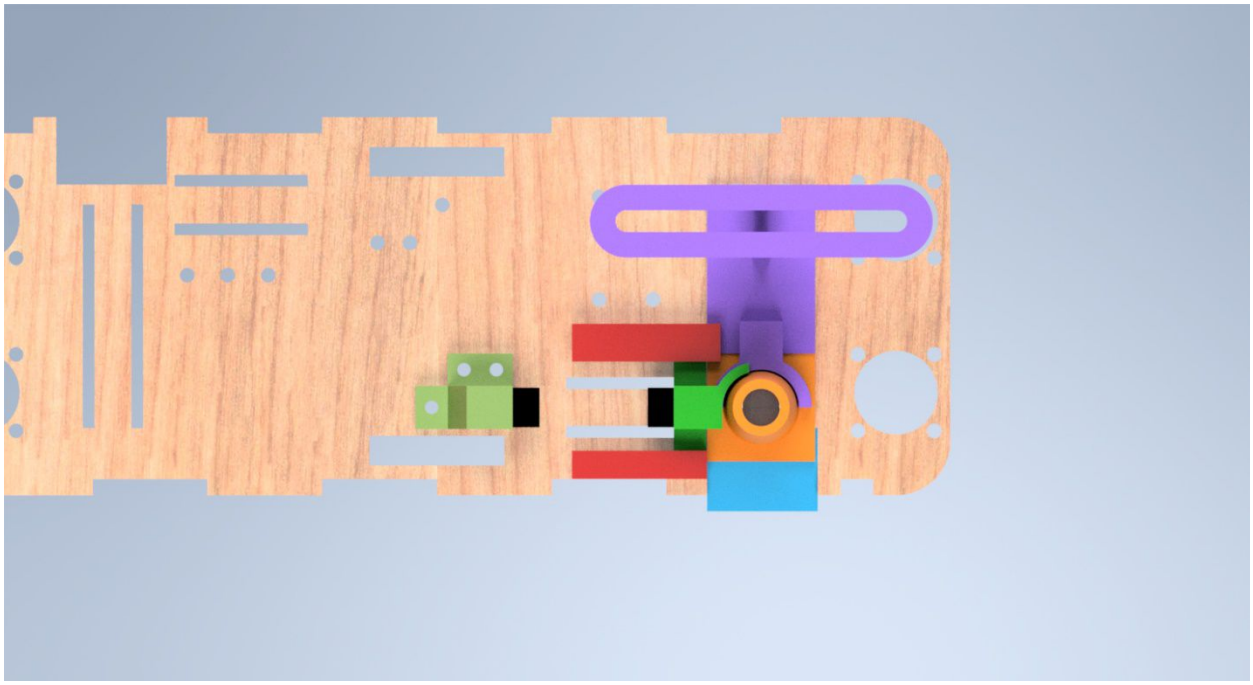


Figure 19. The spring pusher reloading the plunger with another cone

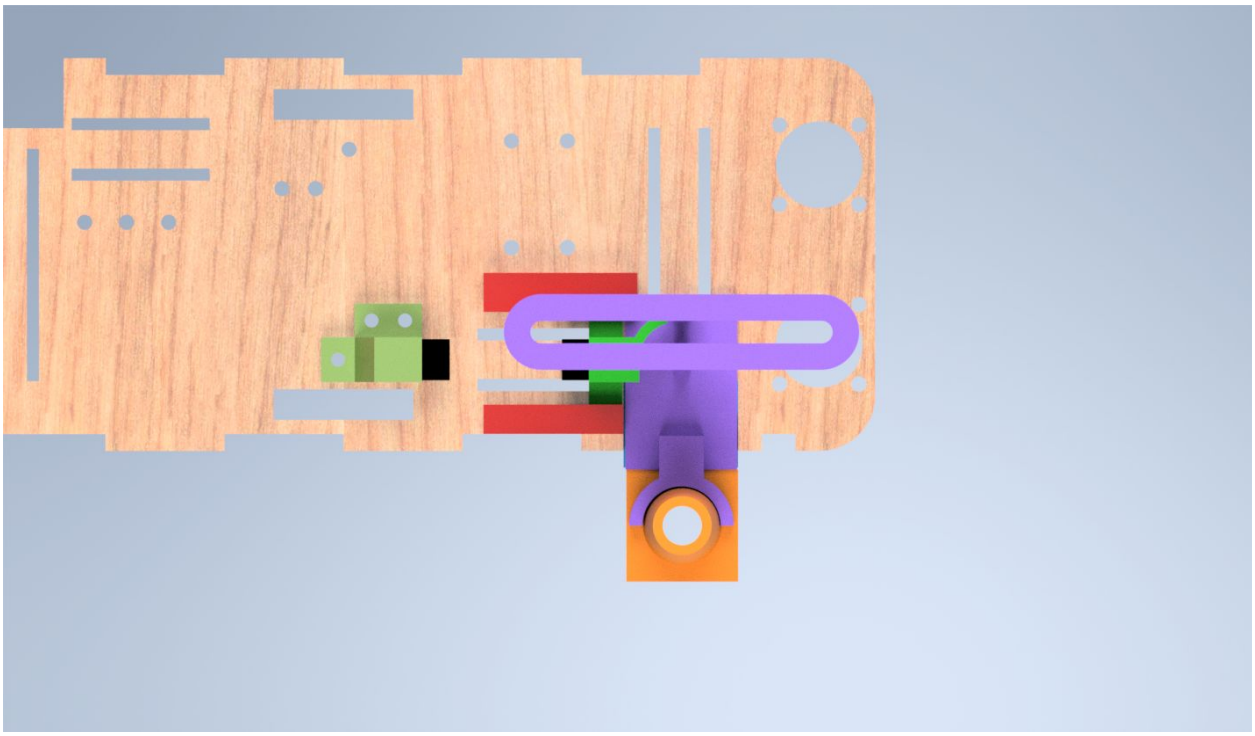


Figure 20. The second cone pushed out by the plunger

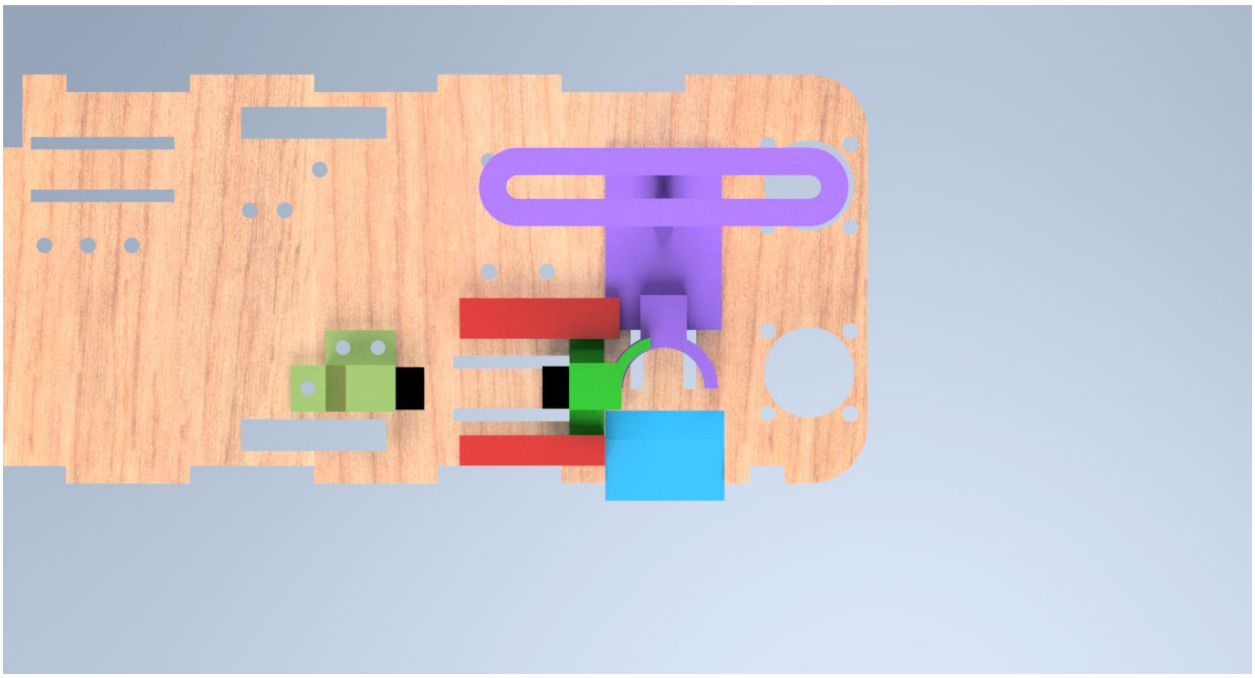


Figure 21. Plan view after both cones have been pushed out

5.5 – Guiderails and Tabs and Slots

To help with the placement of components in the building process and to strengthen the connection, parts were given tabs and respective slots to slot into so that they would have a tight fit. The chassis is held together with Mortensen tenon joints as the top, bottom, left and right parts are individual components. The plunger and pusher also have tabs which fit into a channel that is used as a guiderail to ensure straight movement when they push the cones.

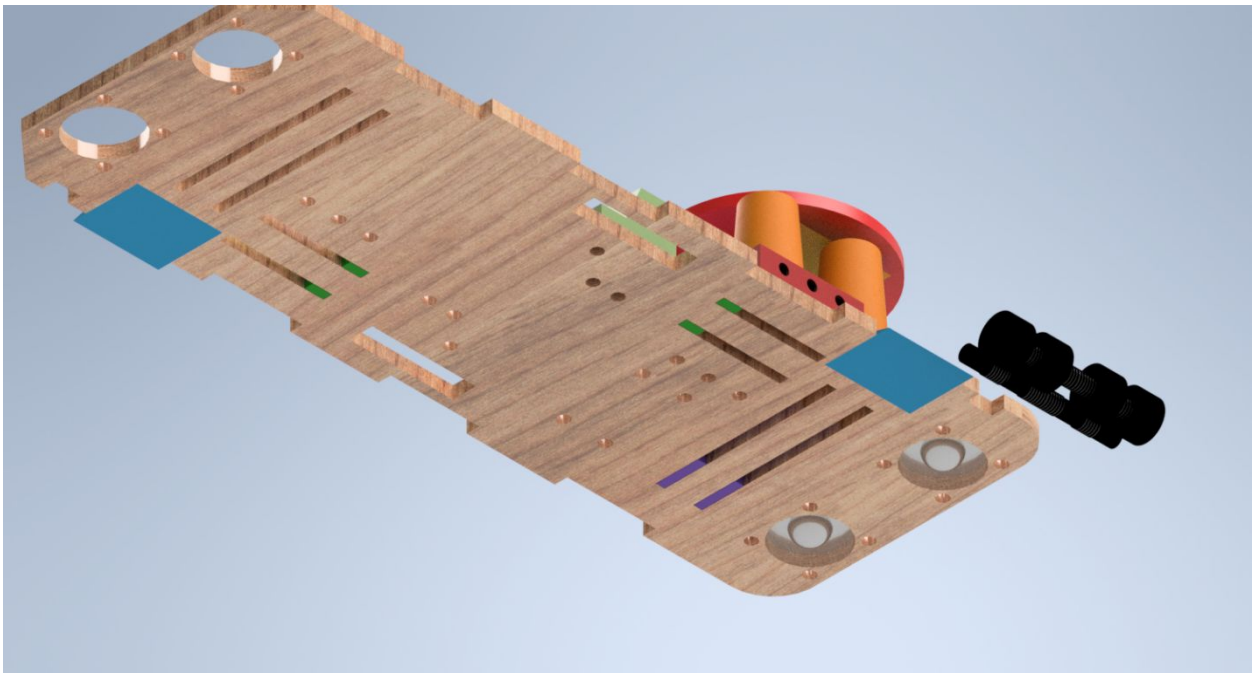


Figure 22. Tabs, slots, and guiderails installed in the chassis floor

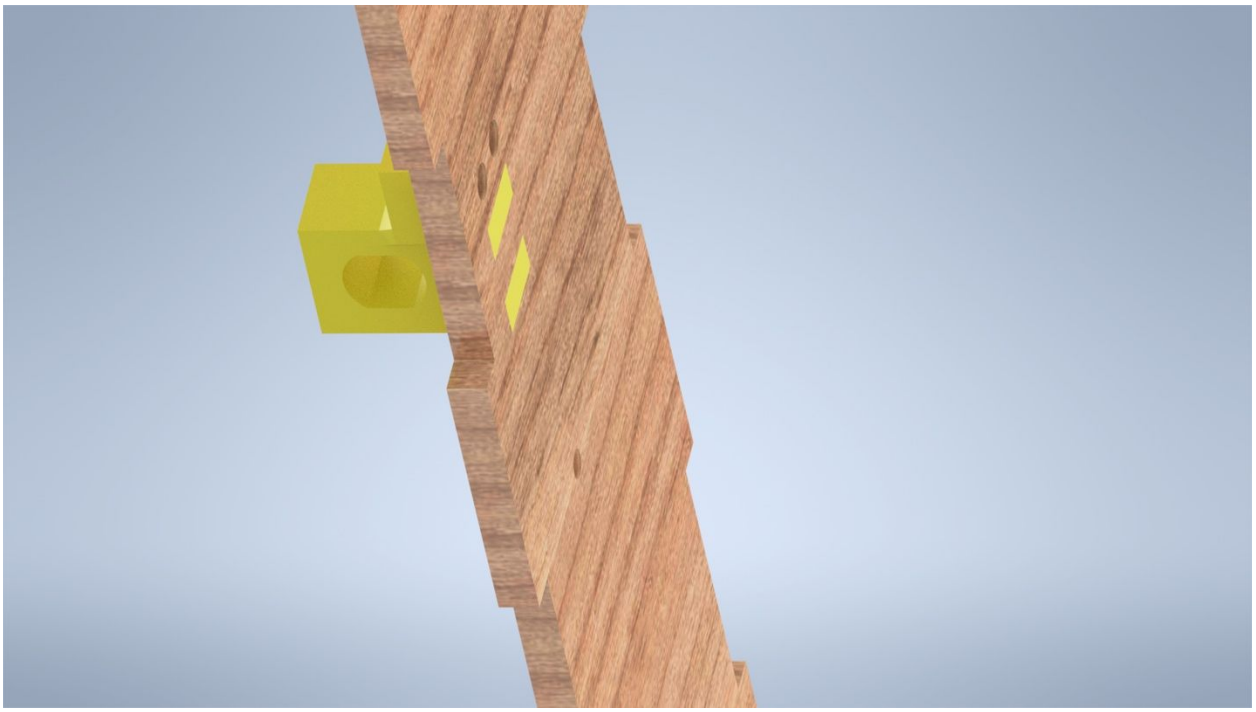


Figure 23. Motor Housing mounted on the Chassis side using tabs and screw holes

5.6 – Spring Retention / Attachment

The springs are attached to the pusher and pusher mount by slotting them into a circular tube. These components (in black) have a diameter that matches the inner diameter of the spring so that the spring can be slotted in. As we do not know what spring we would like to use as we are unsure of the strength required to push the cones, we selected three different springs to use and have created three spring holders to slide into the spring and thread onto the pusher mount and pusher.

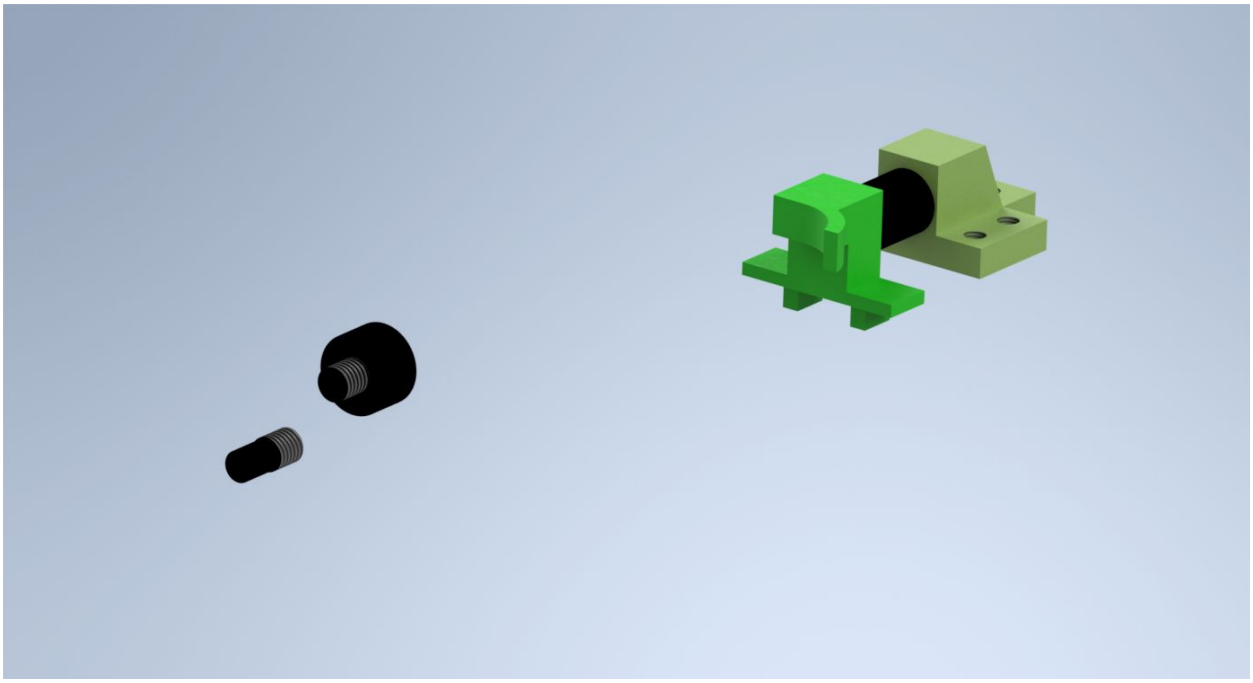


Figure 24. Spring pusher, mount, and holder arrangement

5.7 – Ball Transfer Units

Because the chassis only as space for two wheels, supports would be needed to keep it balanced, originally rollers were going to be used, but we opted to use ready-made Ball Transfer Units (BTU) for smother movement. BTUs have a ball bearing inside that is free to move in all directions, it is frequently used in conveyers and production lines but the inspiration for its use in this application came from the use of a similar part in the LEGO Mindstorms robot set. The robot has two driven wheels and a single ball bearing in an upside-down cup in order to keep it upright. In this application, four BTUs were used to ensure stability in the machine and were mounted onto the body using a 3D printed mount. The end of the mount was also extended to provide a stopping point for the cones pushed by the pusher and spring. The BTUs pressure fit into the mount so this allows some up and down placement adjustment so that the machine is perfectly parallel to the ground.

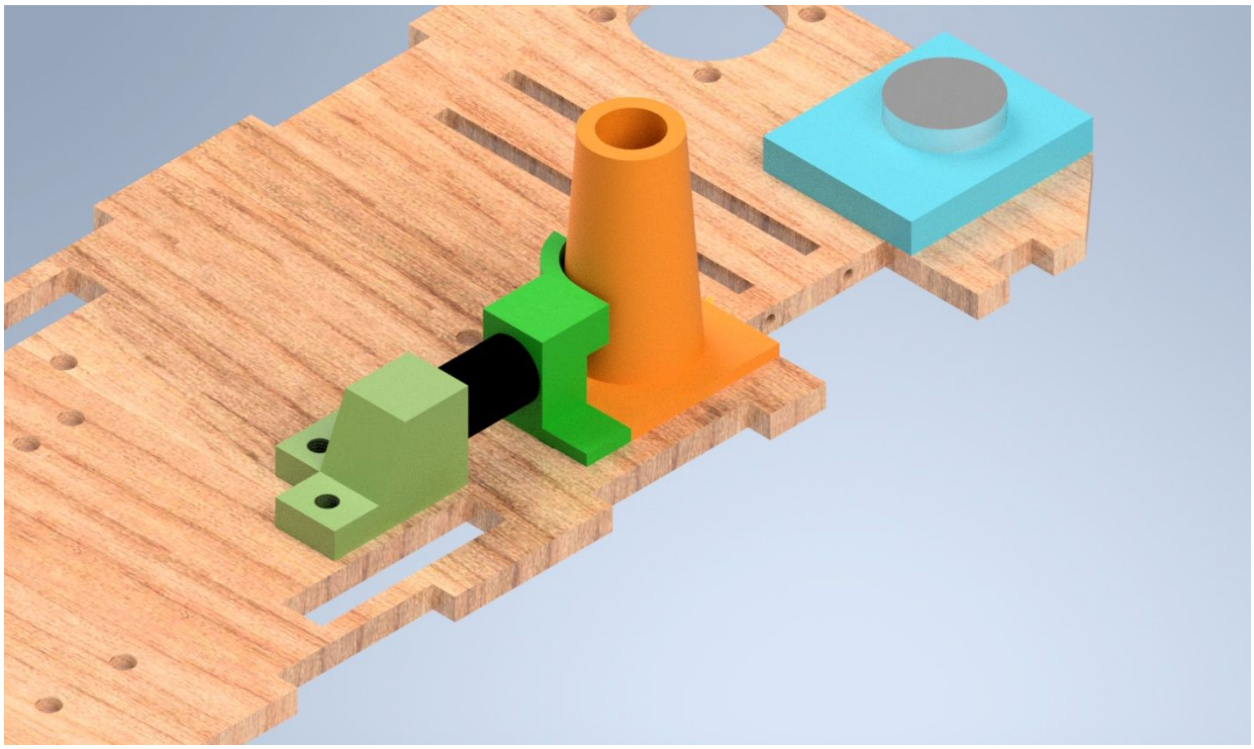


Figure 25. Arrangement of the ball transfer unit

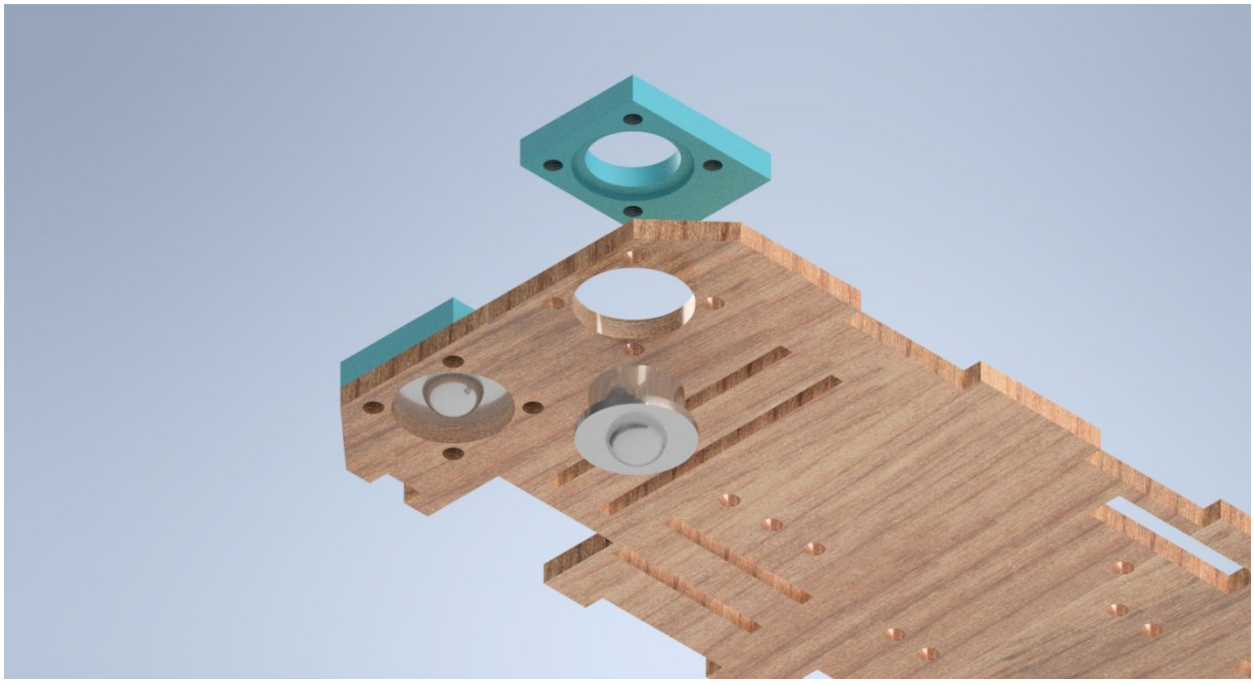


Figure 26. Ball Transfer unit mount

6 - Specification Evaluation

Table 4 shows the evaluation of the final design compared to the design specification in Table 1 to ensure the requirements were successfully met.

Number	Requirement	Evaluation
1	The minimum number of cones that should be dispensed from the vehicle to form a diagonal line of	The final design will place four cones onto the floor forming a double line (parallelogram) pattern of cones
2	The separation distance between the cones in the y-direction should be 200mm	This has been considered and separation distance is adjusted as 200mm
3	The separation distance between the cones in the x-direction should be 200mm	This has been considered and separation distance is adjusted as 200mm
4	Maximum power input of the batteries to the motors	The final design will use less than 9V (DC)
5	The maximum time of the cycle (the time taken to unload the cones)	The final design will place all three cones onto the floor in under 60s
6	The maximum volume of the vehicle-smaller volume is preferable	The volume of the vehicle is 140mm x 380mm x 115mm (width x length x height)
7	The process should be a continuous process	The scotch-yoke mechanism mean that it is a continuous process.
8	The motor must not be subjected to any radial or axial loads	The motors are not subjected to any radial or axial loads

9	The parts should be purchased from approved suppliers or made with CNC or 3D printing	Most of the parts have been 3D printed and laser cut with some of the parts purchased from UoB resources and external suppliers
10	Adhesives cannot be used. Only mechanical fasteners are allowed	No adhesives used in the making of the final design
11	The minimum diameter of the shafts, ensuring they are fully located	The shafts which will be used in the design will have a diameter of 4mm
12	The minimum gear module	All gears have been 3D printed
13	The vehicle can only contain mechanical elements	The final design consists of gears, bevel gears, belt drives and 2 scotch-yoke mechanisms.
14	The maximum number of motors which can be used	One motor is being used in the vehicle to power the vehicle and the scotch-yoke mechanism
15	The maximum cost of parts purchased from the external approved suppliers	The total cost of the vehicle parts came to £74.45 which is under than the maximum of £90,
16	The cones must be on the vehicle before the cycle begins, with no restrictions of their orientation	The cones will be placed on the vehicle before its starts to move

Table 4. Evaluation of Specification

7 - Summary of Meetings

Date	Start	Finish	Present	Absent	Activity
20/03	12:30	14:45	Luca, Elie, James, Pia	Yigit, Advik, Amith	Basic planning and concept making. Exploration of potential mechanics and methods of placing cones.
21/03	14:00	17:00	Luca, Elie, James, Pia, Yigit, Advik	Amith	Individual concept development. Report introduction and product design specification written.
24/03	14:00	15:30	Luca, Elie, James, Yigit, Advik, Amith	Pia	Further individual concept development and development of vehicle using scotch yoke mechanism
27/03	12:15	14:00	Luca, Elie, James, Pia, Yigit, Amith	Advik	Concept evaluation and selection. Development of chosen design. Mechanism selection and calculations
19/04	14:00	15:45	Luca, Elie, Yigit, Amith, Advik	Pia, James	Initial CAD and Report writing
24/04	12:15	14:00	Luca, Elie, James, Pia, Yigit, Advik	Amith	CAD development and Report
28/04	15:00	17:00	Luca, Elie, James, Yigit, Advik	Pia, Amith	CAD development and Report
01/05	10:30	16:00	All		CAD development and Report

Table 5 Summary of Meetings

Table 5 shows a summary of the group meetings. This includes the date, length of time spent working together and the activities that were completed during the meeting.

Table 6 is a Gantt chart, used to plan the project time effectively. An initial timeline was made to allow completion of all tasks before the deadline and to assign roles for each task. The Gantt

chart shows that the completion of many aspects of the project was delayed, resulting in much of the work being completed in the couple weeks before the project deadline.

KEY	
Expected Completion Date	
Actual Completion Date	

ACTION	WEEK (beginning)								ASSIGNED
	20.3	21.3	24.3	27.3	19.4	24.4	28.4	01.5	
PLANNING									
Specification									Luca, James, Elie
Initial Concept									Luca, James, Elie
Individual Concept									All
Concept Evaluation									Elie, James, Yigit
Develop concept									Luca, James, Elie
CAD									
Initial CAD Model									Luca, Pia, Amith, Advik
Development of									Luca, Amith, Advik
Part Drawings									Pia, Advik, Amith
Part List									Amith
3D printing files									Amith, Advik
Laser cutting files									Amith

REPORT									
Introduction									Pia, James, Elie, Yigit
Specification									Luca, James, Elie
Concept drawings									All
Concept Selection									James, Elie, Yigit
Evaluation									Luca, James, Elie, Yigit
Design									Luca, James, Elie, Yigit
Summary of									All
Order forms									Yigit, Luca
Mechanical									James, Luca
Cost Calculations									Yigit
Conclusion									Luca
Summary									Pia, Elie, Yigit, James
Appendices									Pia and Advik
Final Review									Luca, James, Elie, Amith

Table 5 Summary of Meetings

8 - Order Forms and Cost Calculations

8.1 – Budget for external suppliers

The parts purchased from external suppliers cost £73.483.

Qty	Supplier Part no.	Supplier	Description	Unit Price £	Total £
2	121-242	RS PRO	RS PRO Alloy Steel Compression Spring, 110mm x 21.6mm, 0.99N/mm	£1.972	£3.944
2	121-185	RS PRO	RS PRO Alloy Steel Compression Spring, 94mm x 17.25mm, 0.72N/mm	£1.93	£3.86
2	751-512	RS PRO	RS PRO Alloy Steel Compression Spring, 80.5mm x 8.63mm, 0.17N/mm	£0.918	£1.836
1	1451-252	Technobots	Pololu 380:1 Micro Metal Gearmotor	£21.73	£21.73
1	4607-072	Technobots	Model Sleeve Reducer 4/3mm	£0.155	£0.155
1	4600-060	Technobots	Plastic Bevel Gear Pair 4mm	£3.28	£3.28
4	4609-040	Technobots	Zinc Collets 4mm pk/4	£1.46	£5.84
1	4357-306	Technobots	Pk 10 Grub Screw (Socket Set) M3 x 6mm	£0.55	£0.55
3	6001-042	Technobots	20 Tooth 8mm Bore 2mm Pitch GT2 Pulley with for 6mm Belts	£1.64	£4.92
1	6001-448	Technobots	6mm Wide GT2 / 2GT 2mm Pitch Rounded Tooth Profile 336mm Closed Belt	£2.84	£2.84
1	6001-442	Technobots	6mm Wide GT2 / 2GT 2mm Pitch Rounded Tooth Profile 320mm Closed Belt	£2.84	£2.84
4	687-821	RS PRO	ALWAYSE Ball Transfer Unit with 15mm diameter Steel ball	£5	£20
1	1063-072	Technobots	4 x AAA Enclosed Battery Box with Leads	£0.79	£0.79
4	270-552	RS PRO	3mm Diameter Plain Steel Parallel Dowel Pin 12mm Long	£0.202	£0.808
Overall Cost		£73.483			

Table 6 Approved external suppliers order form

8.2 – Uni of Bath resources

The parts purchased from the Uni of Bath resource list cost a total of £1.7212.

Qty	Supplier Part no.	Description	Unit Price £	Total £
1	SB004300	UOB 4mm Steel Shaft (150 mm) (Modified	£2 per 300mm	£1
1	SB004300	UOB 4mm Steel Shaft (60 mm) (Modified	£2 per 300mm	£0.4
1	SB004300	UOB 4mm Steel Shaft (20 mm)	£2 per 300mm	£0.13
40	N/A	Countersunk M5 Bolt 12mm (Black Metal)	N/A	N/A
4	209-6851	Circlips 6mm x 0.8mm height	£0.0295	£0.118
3	289-196	Circlips 4mm x 0.6mm height	£0.0244	£0.0732
4	BATAA	AA Battery	N/A	N/A
Overall		£1.7212		

Table 7 University of Bath order form

8.3 – 3D Printer Cost

The overall cost of 3D-printed parts is £92.41, this includes the £2.10 set-up charge per part.

Quantity	Part Number	Description	Volume (mm³)	Cost – 3D
2	G48001	Pusher	10381.011	£2.08
1	G48002	Battery Housing	29290.221	£2.93
1	G48007	Motor Housing	7505.674	£0.75
1	G48008	Motor Pulley	4300.667	£0.43
2	G48009	Scotch Yoke	32748.747	£6.56
1	G48010	Shaft	1879.458	£0.19
1	G48011	Shorter Shaft	742.987	£0.07
1	G48012	Smallest Shaft	251.327	£0.03
1	G48013	Scotch Yoke Gear	9023.895	£0.90
1	G48014	Mutilated Gear	39633.774	£3.96
1	G48015	Wheel Gear	2231.734	£0.22
1	G48016	Mutilated Gear	39611.962	£3.96
1	G48017	Scotch Yoke Gear	10690.297	£1.07

4	G48018	Ball Transfer Unit	10906.350	£4.36
2	G48019	Centre Cone Rail	6342.666	£1.27
2	G48020	Plunger	29967.551	£6
2	G48021	Back Pusher	6570.365	£1.32
2	G48022	Ramp	4956.795	£1
4	G48023	Thread In Spring	3061.168	£1.24
4	G48024	Thread In Spring	828.728	£0.33
4	G48025	Thread In Spring	2099.291	£0.84
2	G48026	Wall Cone	4154.081	£0.83
2	G48027	Wheel	18832.335	£3.77
Overall Cost		£92.41		

Table 8 3D Printed Parts Cost

8.4 – Laser Cutter Cost

The overall cost of the laser-cut parts is £24.91, This includes the £15 set-up charge for all parts.

Part Number	Description	Area / mm²	Quantity	Thickness (mm)	Cost
G48003	Chassis (Bottom)	104799.520	1	6	£3.4
G48004	Chassis (Left Panel)	51913.794	1	6	£1.68
G48005	Chassis (Right Panel)	51657.482	1	6	£1.67
G48006	Chassis (Top)	97545.405	1	6	£3.16
Overall Cost		£24.91			

Table 9 Laser Cut Parts Cost

8.5 – Student Labour

The labour cost amounted to £835

Task	Cost per Hour	Hours Spent	Total Cost
Concept Drawings	£7	15	£105
Design Development	£8	10	£80
CAD Drawings	£15	30	£450
Report	£10	20	£200
Overall Cost		£835	

Table 10 Student Labour Cost

8.6 – Overall Cost

From summing all the costs, the overall cost amounts to £1027.77

Expense	Cost
External Suppliers Costs	£73.483
University of Bath Resources Costs	£1.7212
CADCAM Costs	£117.32
Student Labour Costs	£835
Overall cost	**£1,027.5242**

Table 11 Overall Cost

8.7 – Cost Evaluation

A big number of parts are 3D printed and laser cut. This procures an important decrease in money spent on external suppliers. The budget for external suppliers was determined as £90. The overall cost is high due to the money spent on 3D-printed and laser-cut parts.

80 per cent of the overall cost is student labour. Subtraction of this cost leaves £192.5242 expense.

9 - Conclusion

A cone unloader prototype is designed and modelled to enhance the working conditions of road workers. In this design, magazine-like contraptions with spring pushers are designed to store cones and push them to the ejection point. Two spring and scotch yoke mechanisms work harmoniously to unload cones onto the road in adjusted measurements. The rail mechanism is designed to prevent cones to tip over due to the push of the spring pusher and plunger. The vehicle is adjusted with special 3D-printed gears to stop while unloading cones onto the road. All essential elements are used to design this prototype.

9.1 - Improvements to process

The project was generally handled efficiently and was submitted in time for the deadline, the budget was respected, there were no major problems with any of the components of the assignment and on paper, the machine works and is able to dispense the cones in the correct order and arrangement. But this does not mean that some aspects of the could have been handled better.

The planning and allocation of roles could be improved upon for a possible next project. At times, not every member was aware of what they should have been doing, or more often, how to do it, there were also issues with work submission as deadlines were not explicitly specified or stated. This required the person assigning the roles to re-explain, in more detail, what should have been done. This resulted in time being wasted as tasks were not being done properly. A possible way

to fix this would be to better improve communication between team members and to make sure that everyone knew what they had to do, how to do it and by when to do it.

But the confusion with roles and tasks to perform might also have been due to the complexity of the design. As pointed out by other groups, the design chosen and developed appears to be complex and in need of very precise manufacturing, something 3D printers and laser cutters may not always be able to deliver. As the design has not been built nor tested yet, the hindrance of its performance due complexity remains to be seen. But its complexity could pose as an issue, so for a possible next project, a simpler approach may be preferred, as stated by businessman, investor, and engineer Elon Musk: “The best part is no part, the best process is no process. It weighs nothing, it costs nothing and therefore cannot fail.”, this kind of approach to design has its merits and could be used as a basis of design for a future project.

As for improvements to the current design, it is hard to know exactly what may go wrong as it has not been tested yet. Possible areas of failure could be the gear ratios as the dimensions of the gears resulted in being quite awkward towards the end of the project, this would result in the cones not being placed in the exact position to create the desired pattern. 3D printed threads and tapped holes could also not mesh properly in the spring holders, pusher mount and pusher components, as well as all parts fastened with the M5 countersunk bolt, this could also be another point of failure or incompatibility of the design that could be fixed by simply gluing the parts in place, but the design brief specifically prohibits the use of glue so that fix would not work in this scenario. Some 3D printed and laser cut parts could break due to stresses in the machine or simply because of their design and fragility as with CAD software, one often loses the sense of scale and parts may be designed to be very small and by consequence very fragile. Parts that may be at risk of breaking are the Scotch Yoke plunger, the bottom and sides of the chassis and motor pulley bottom. This could incentivise the use of more ready-made parts, but their use implies another set of problems and complications.

Three suppliers were used for the purchase of ready-made parts: RS Components, the University of Bath and Technobots. While RS Components had detailed drawings of their parts as well as detailed specifications, and the University had its in-house stock supply of parts on hand to be measured and test fitted, Technobots had less than satisfactory drawings of their parts, if at all given as many parts did not have drawings, and very vague specifications of their components. Whether they cause an issue, that remains to be seen, but in the future the use of Technobots

parts is not recommended, not based on their quality (which also remains to be seen), but based on the fact that precise dimensions and tolerances are not specified. Components that had already been bought had to be manually measured to be able to ensure a good fit. Availability and stock or components were also an issue, which is why the gears were opted to be made in house with 3D printers.

Overall, the group worked efficiently, and no major complications arose in the design process of the machine, a few setbacks occurred during the process and a more even participation of all members would have led to a more efficiently handled project. But the goal of designing a cone dispensing machine and producing a report along with part drawings and files for all components was achieved successfully within the given time frame.

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Appendix A:

Assembly Drawings with Parts List Table

Reference	Part Number	Description	Quantity
1	G48003	Chassis (Bottom)	1
2	G48004	Chassis (Left Panel)	1
3	G48005	Chassis (Right Panel)	1
4	G48022	Ramp	2
5	N/A	Cone (V1) (Provided by UOB)	4
6	G48006	Chassis (Top)	1
7	G48026	Wall Cone Rail	2
8	G48019	Center Cone Rail	2
9	G48001	Pusher	2
10	G48021	Back Pusher	2
11	G48018	Ball Transfer Unit Mount	4
12	N/A	BTU Mockup (Provided by RSC)	4
13	N/A	Floor (Provided by UOB)	1
14	G48010	Shaft	1
15	G48026	Wheel	2
16	N/A	Collet Mockup (Provided by TNB)	4
17	G48009	Scotch Yoke	2
18	G48017	Scotch Yoke Gear (Top Mech)	1
19	G48025	Thread In Spring Attachment	4
20	G48023	Thread In Spring Attachment Biggest Spring	4
21	G48023	Thread In Spring Attachment Smallest Spring	4
22	N/A	Universal Coupling 6mm Mockup (Provided by TNB)	2
23	N/A	Universal Coupling 4mm Mockup (Provided by TNB)	3
24	N/A	Bevel Gear 1 (Provided by TNB)	1
25	N/A	Bevel Gear 2 (Provided by TNB)	1
26	G48011	Shorter Shaft	1
27	N/A	Motor Mockup (Provided by TNB)	1
28	N/A	Universal Joint Mockup (Provided by TNB)	1
29	G48007	Motor Housing	1
30	G48012	Smallest Shaft	1
31	G48008	Motor Pulley Bottom	1
32	N/A	Synchronous Belt (Provided by TNB)	1
33	N/A	Synchronous Pulley 1 (Provided by TNB)	1

34	N/A	Synchronous Pulley 3 (Provided by TNB)	1
35	N/A	Synchronous Pulley 2 (Provided by TNB)	1
36	N/A	Battery Box Mockup (Provided by RSC)	1
37	G48002	Battery Housing	1
38	G48020	Plunger	2
39	G48013	Scotch Yoke Gear (Bottom Mech)	1
40	G48014	Mutilated Gear (Bottom Mech)	1
41	G48016	Mutilated Gear (Top Mech)	1
42	G48015	Wheel Gear	1

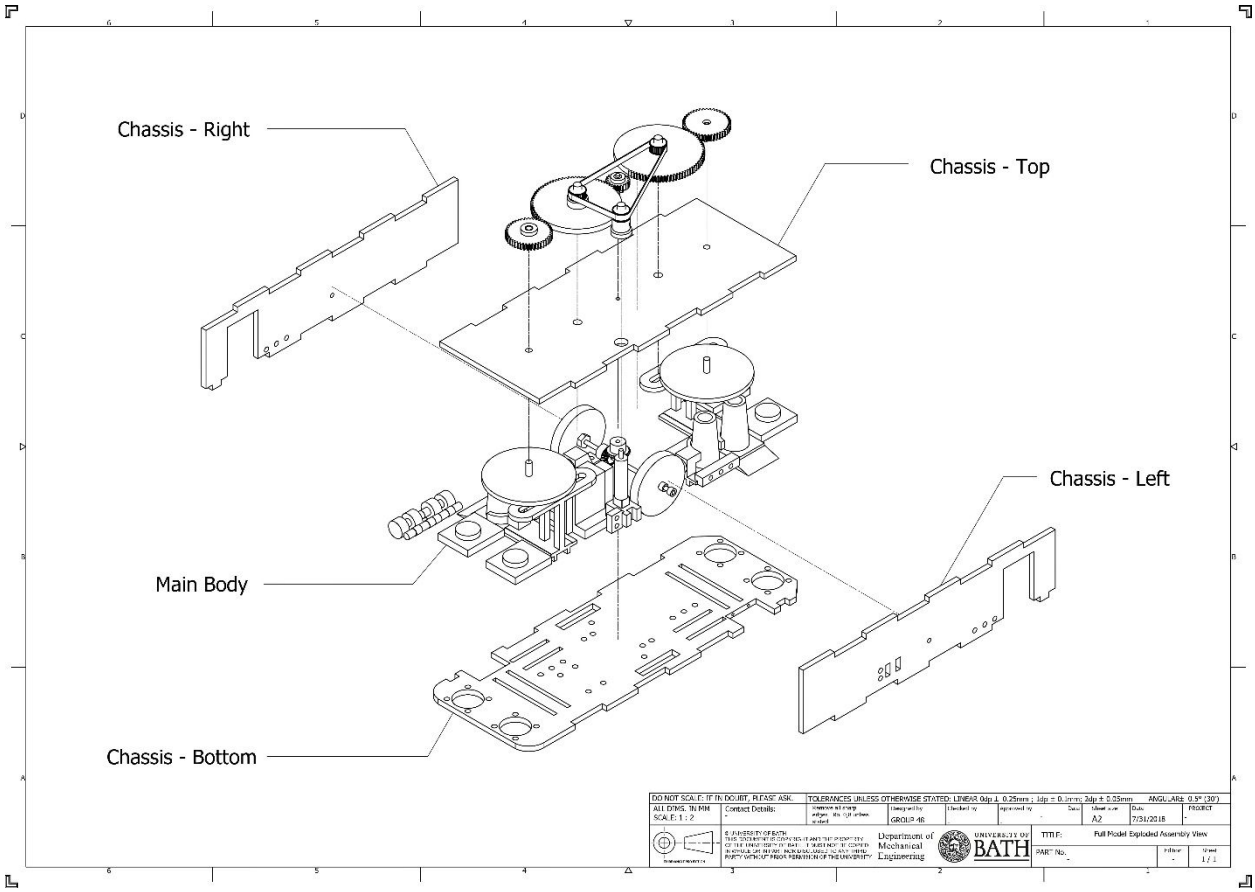


Figure 27. Full Model Exploded View

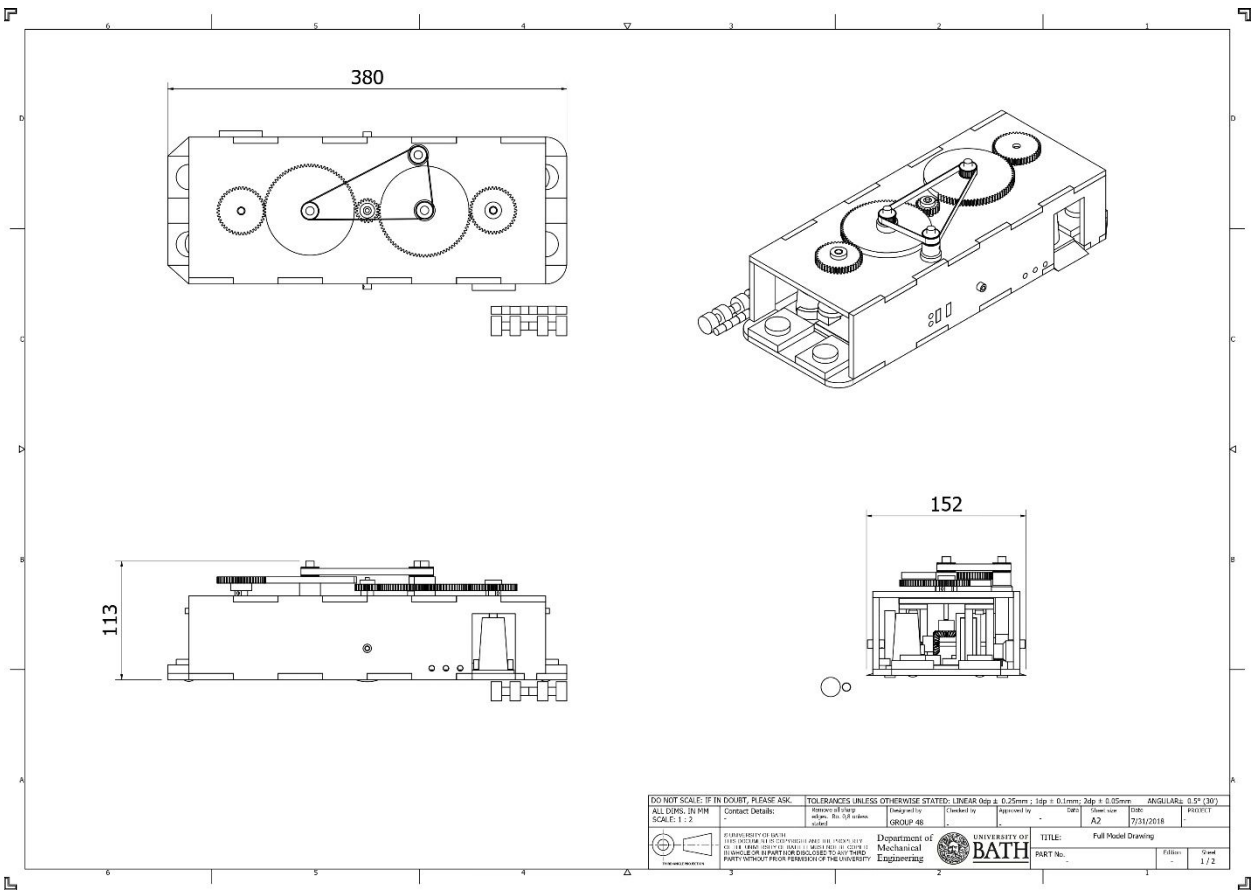


Figure 28. Full Model Assembly

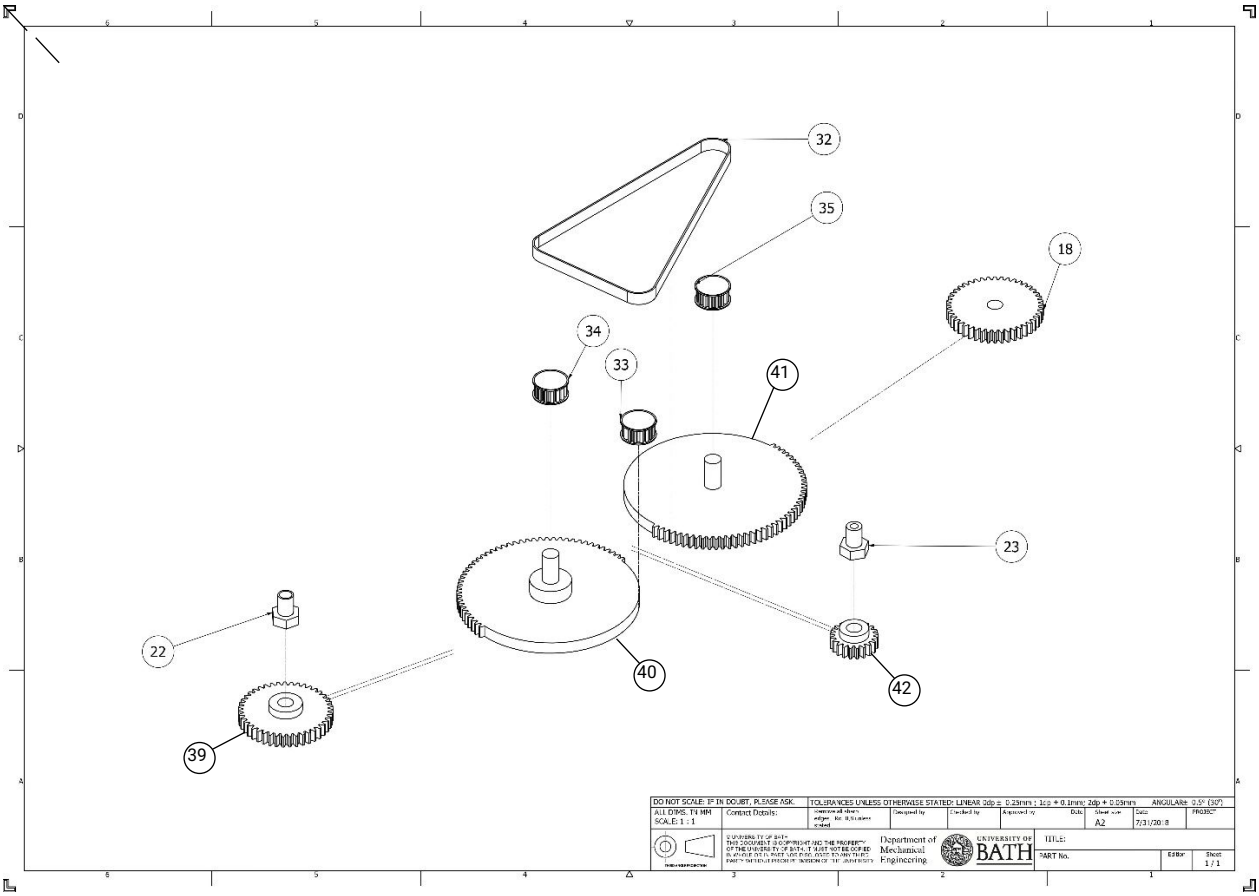


Figure 29. Gear Mechanism Assembly

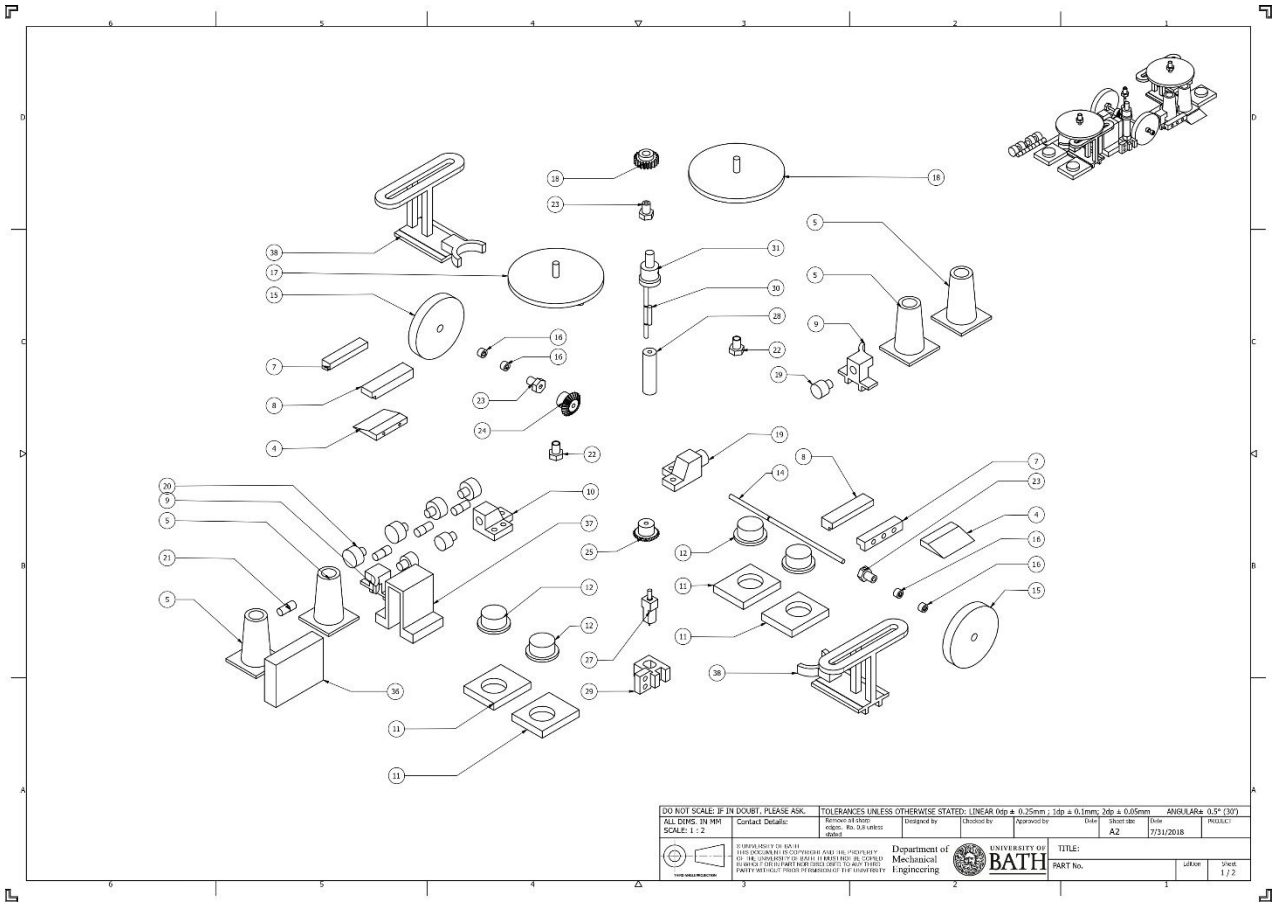


Figure 30. Main Body Exploded View

Appendix B: Part Drawings

The parts drawings present only display the custom-made parts (those with part numbers), purchased parts such as: The Traffic Cone, Ball Transfer Unit, Collet, Universal Couplings and Joints, Synchronous Belt and Pullies, Bevel Gears and Motor and Battery Box are not shown.

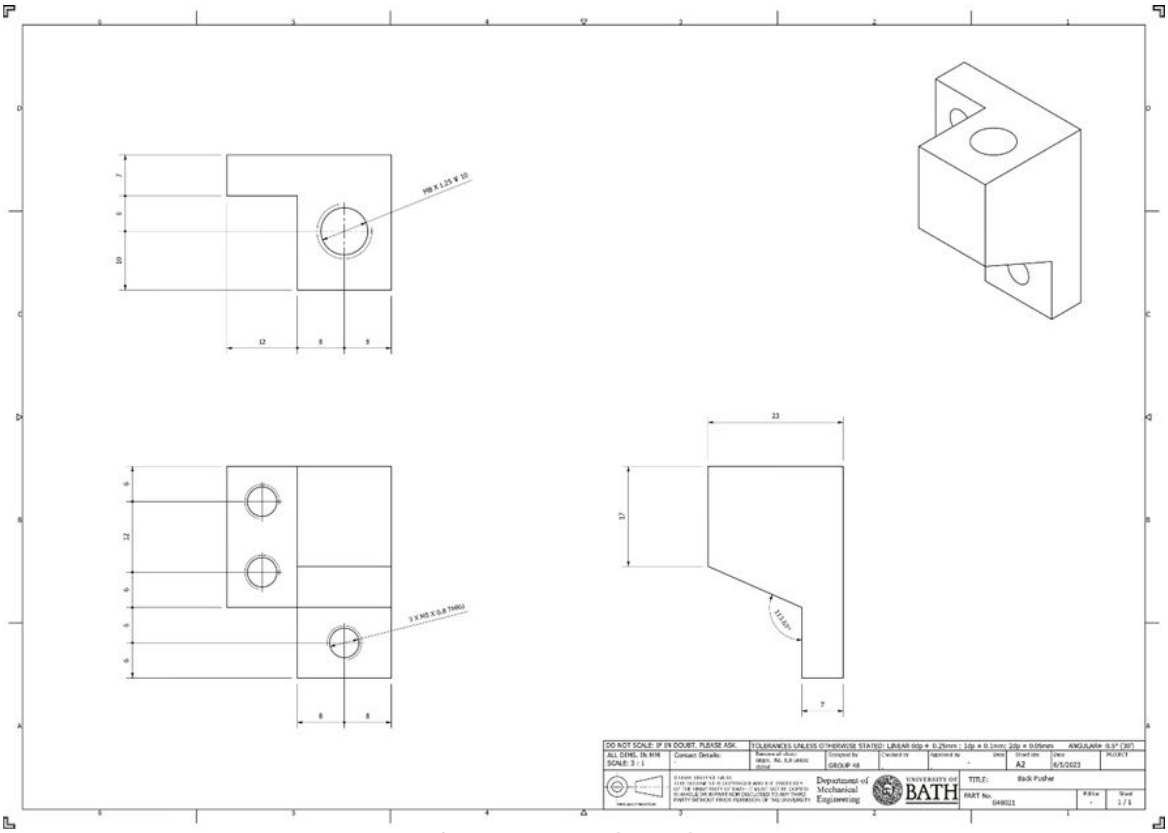


Figure 31. Back Pusher

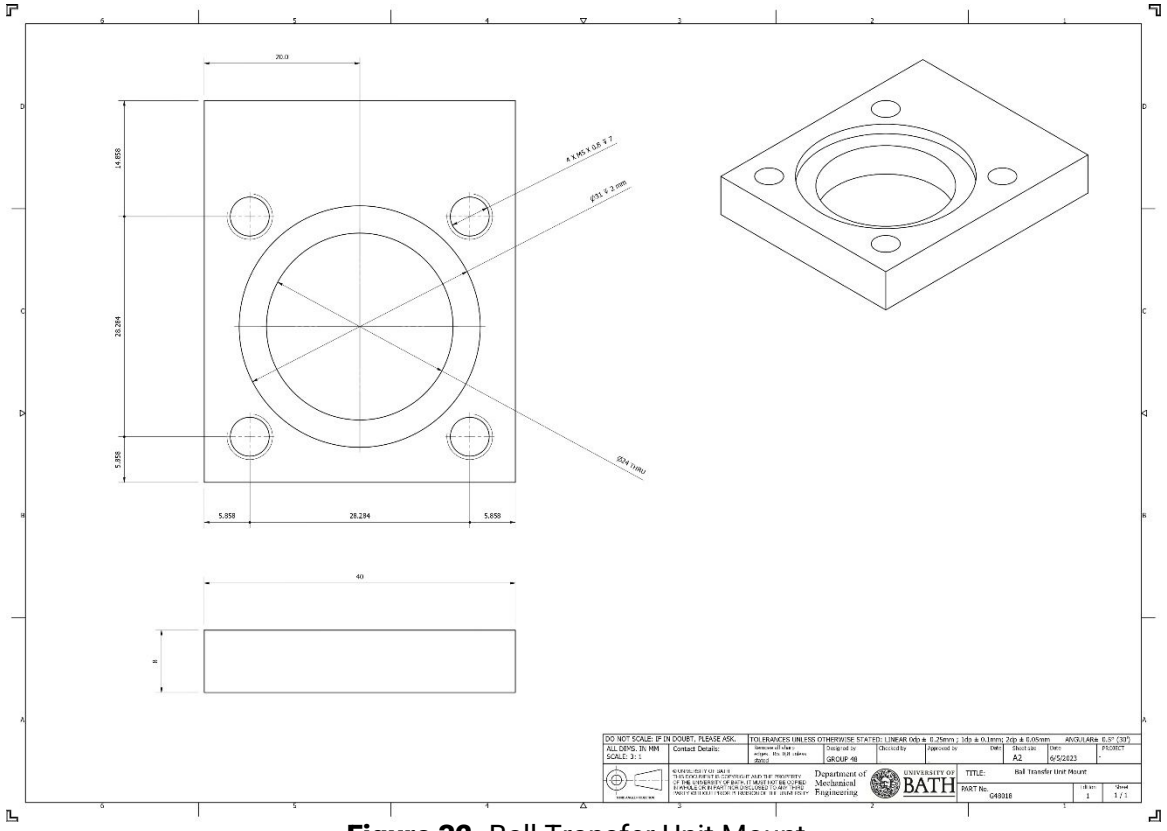


Figure 32. Ball Transfer Unit Mount

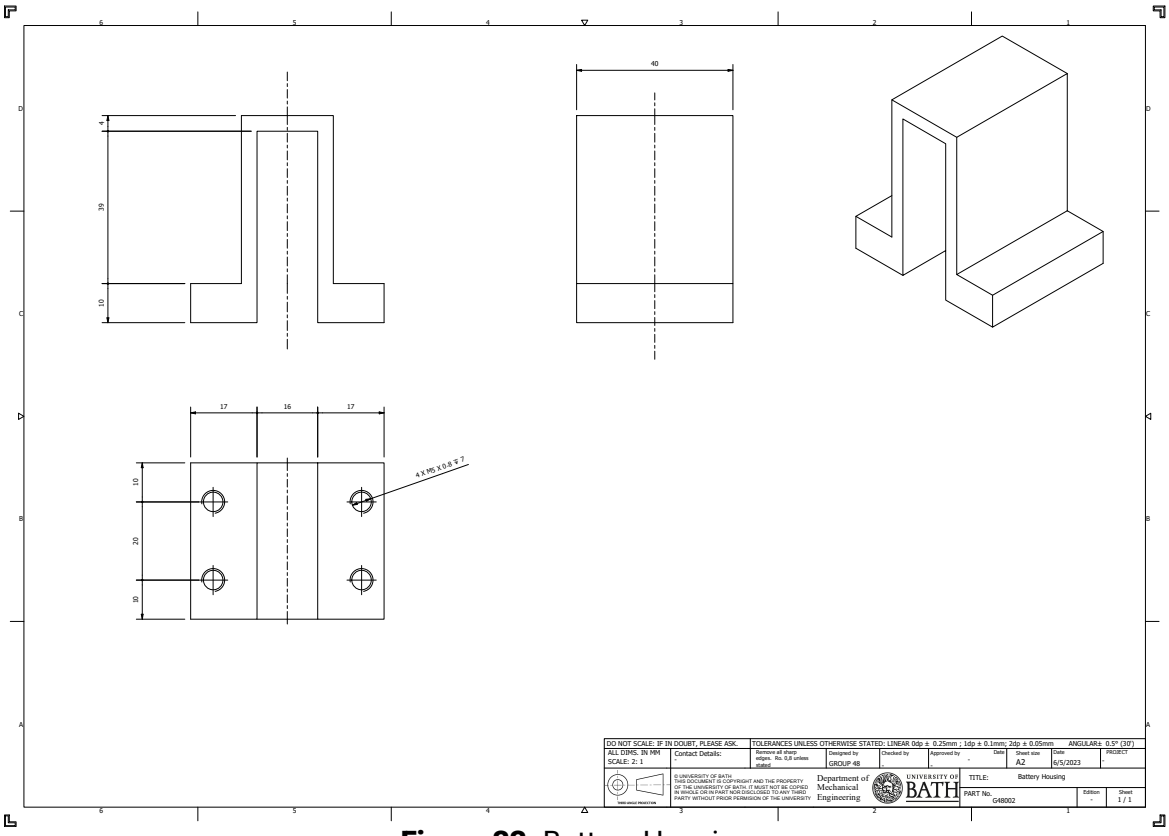


Figure 33. Battery Housing

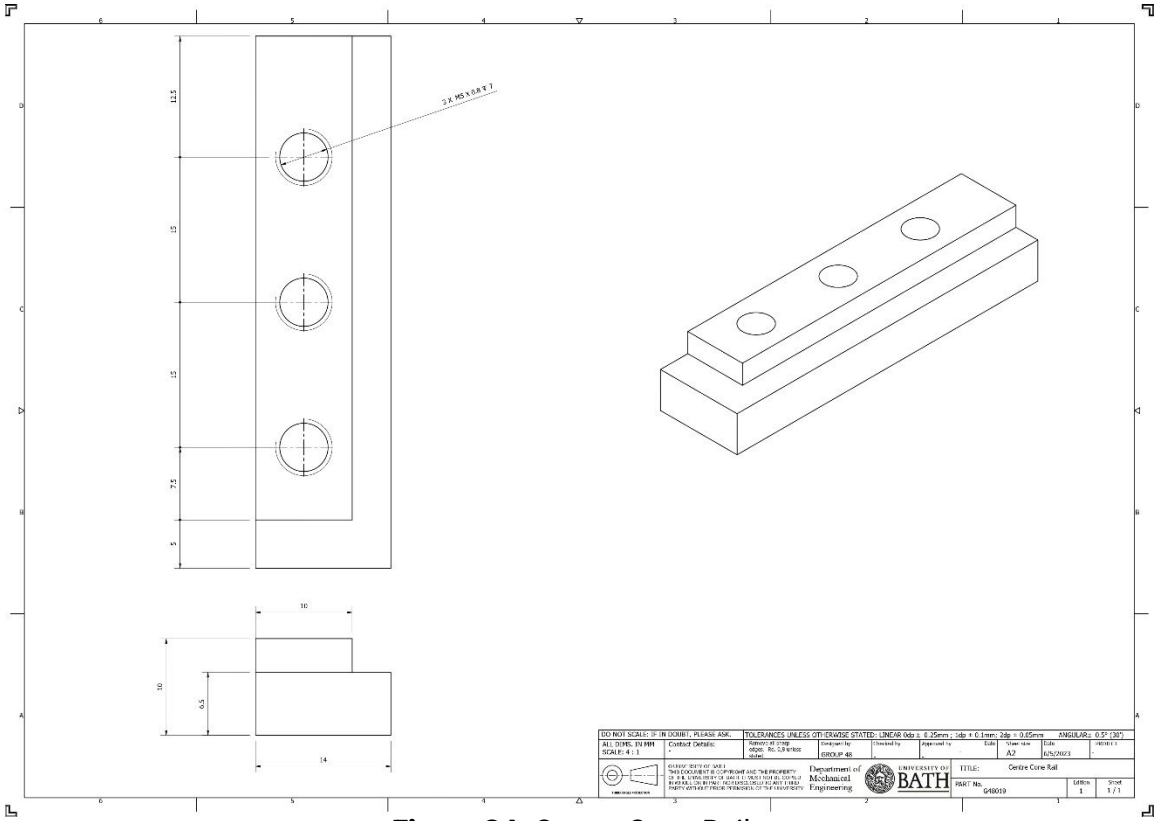


Figure 34. Centre Cone Rail

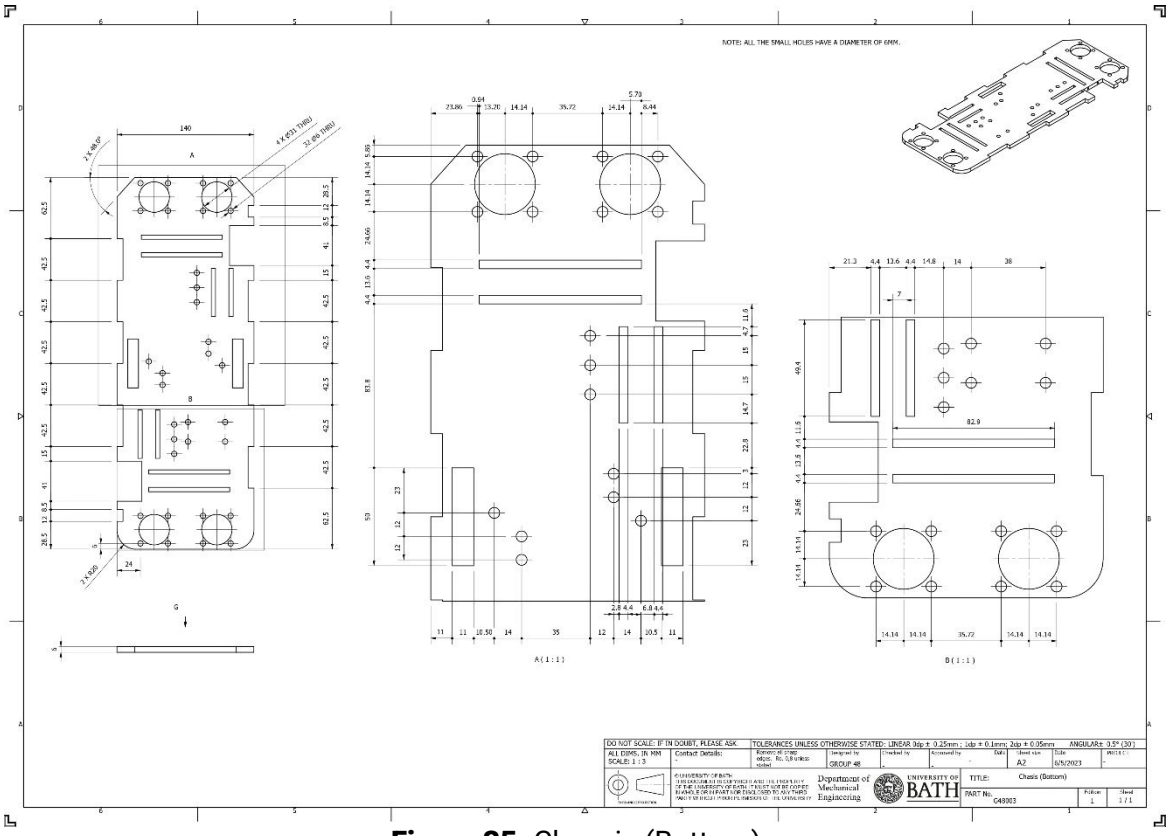


Figure 35. Chassis (Bottom)

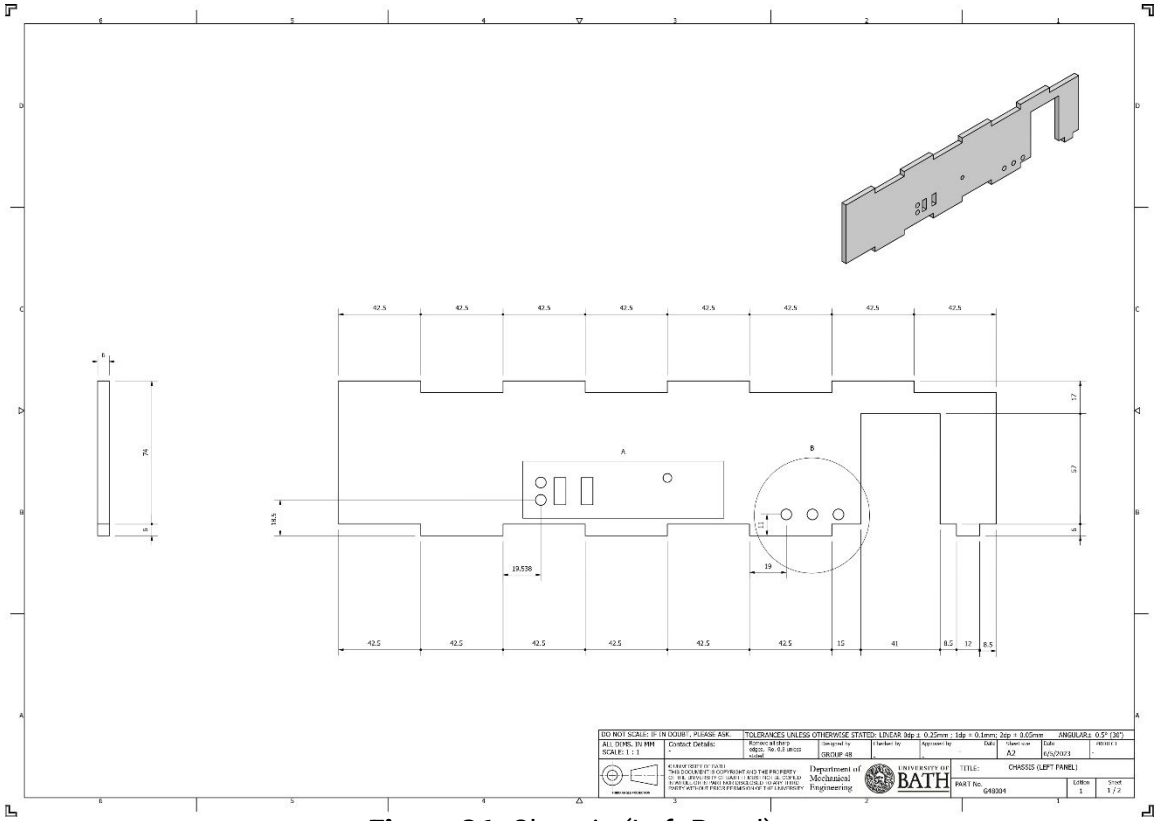


Figure 36. Chassis (Left Panel)

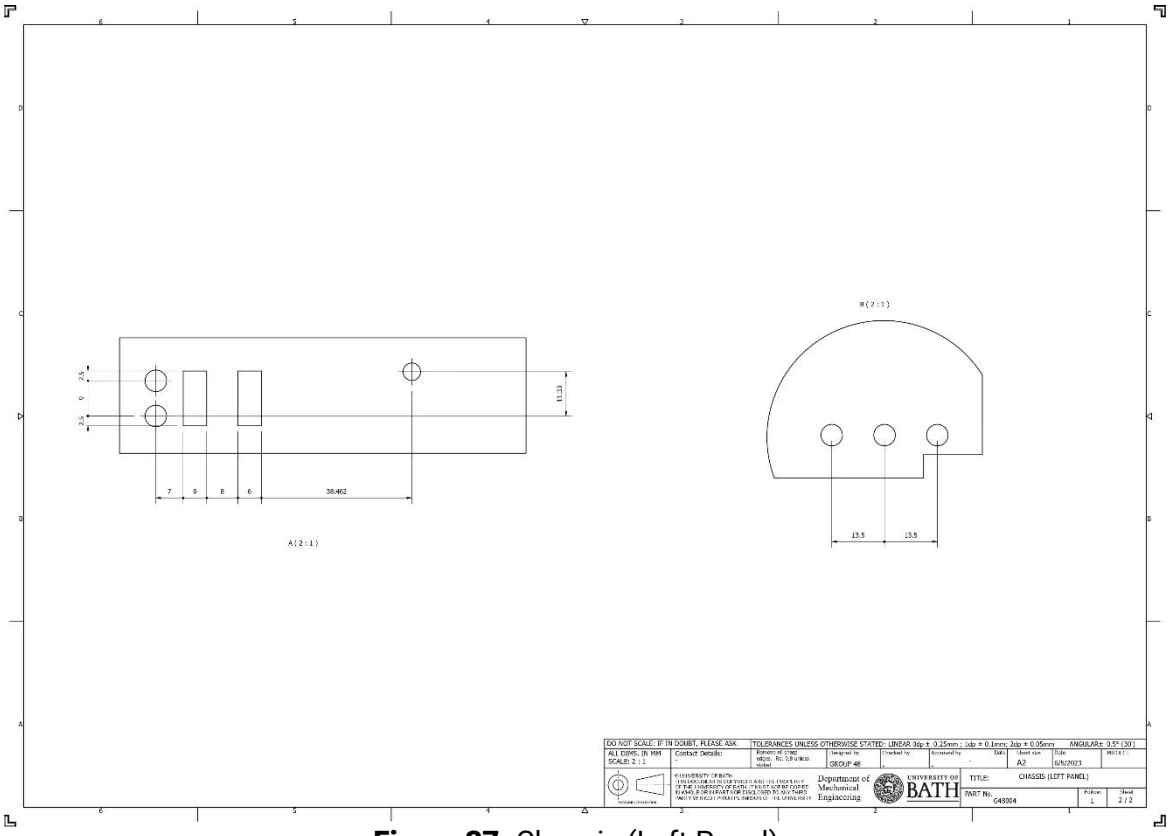


Figure 37. Chassis (Left Panel)

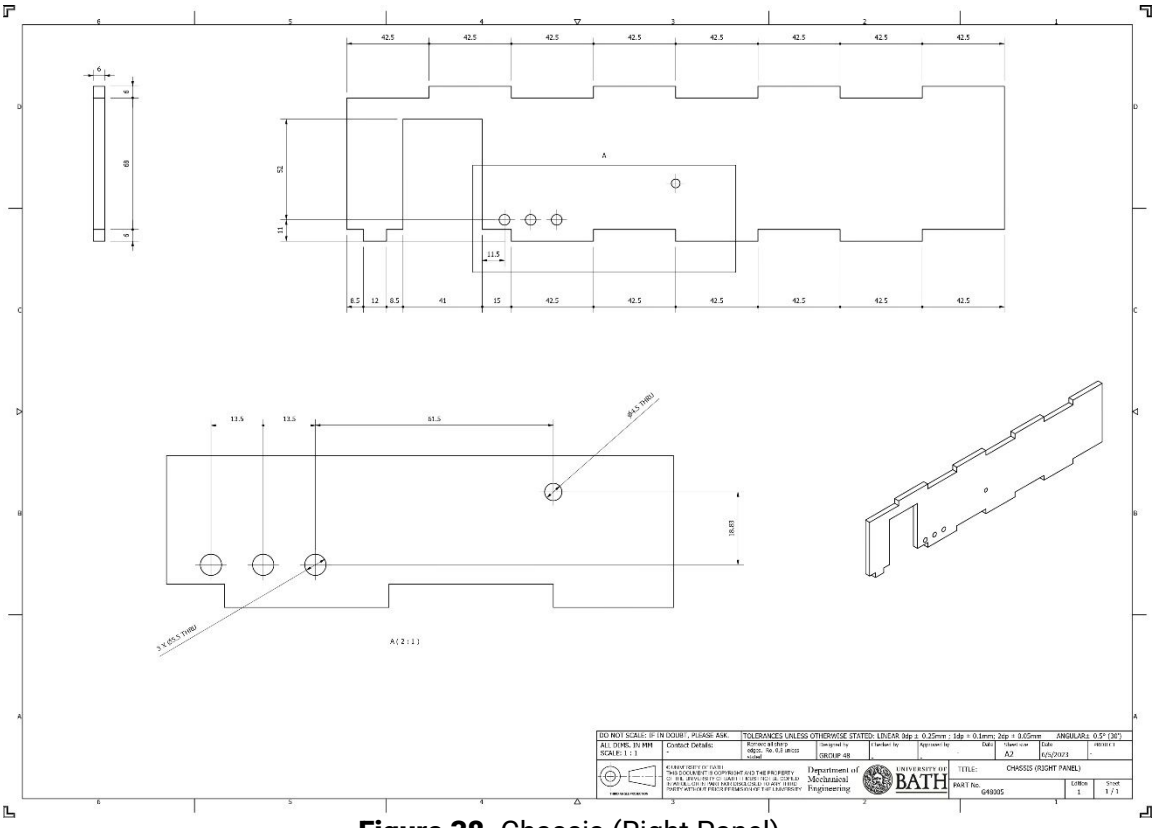


Figure 38. Chassis (Right Panel)

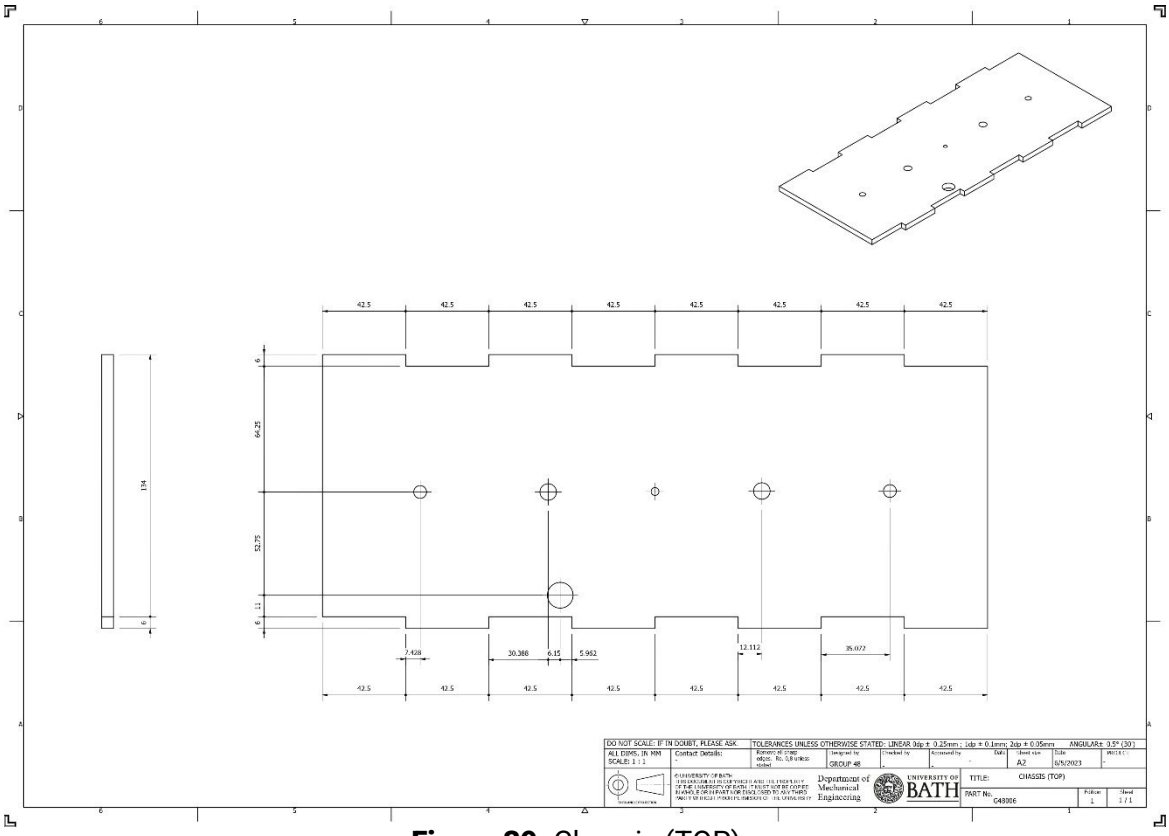


Figure 39. Chassis (TOP)

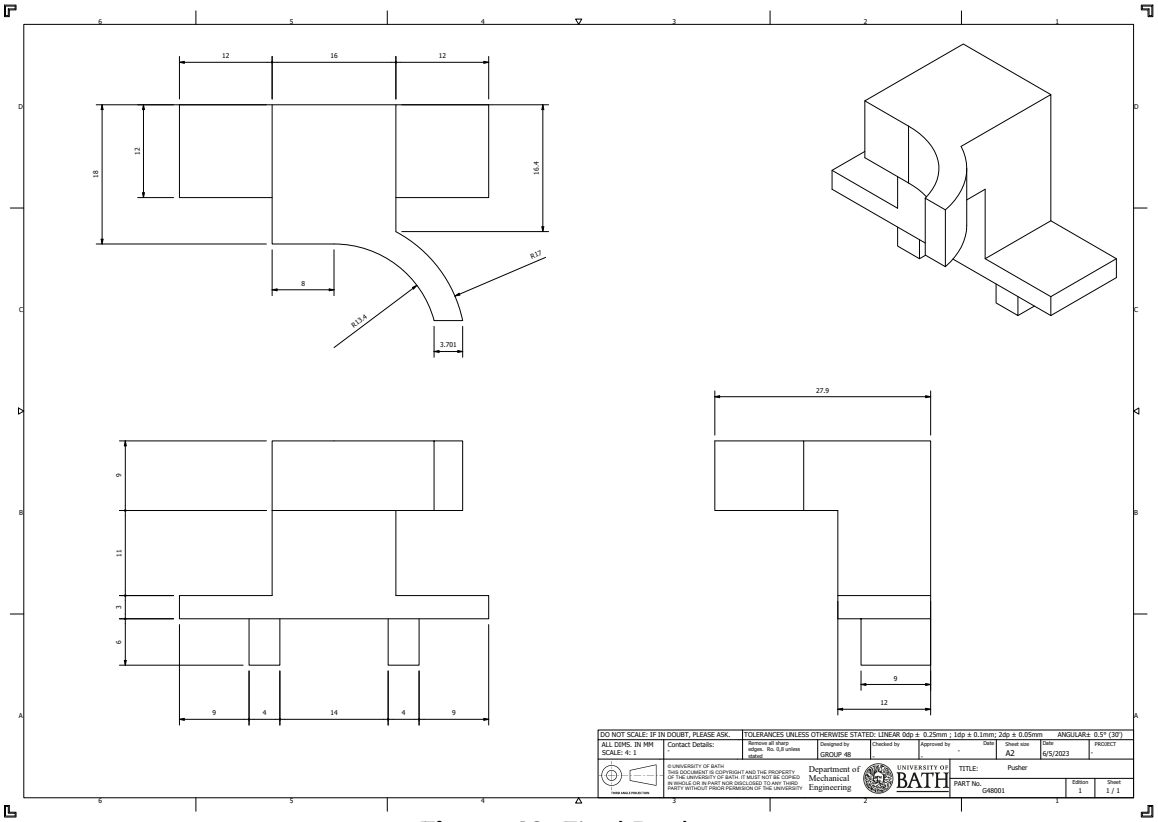


Figure 40. Final Pusher

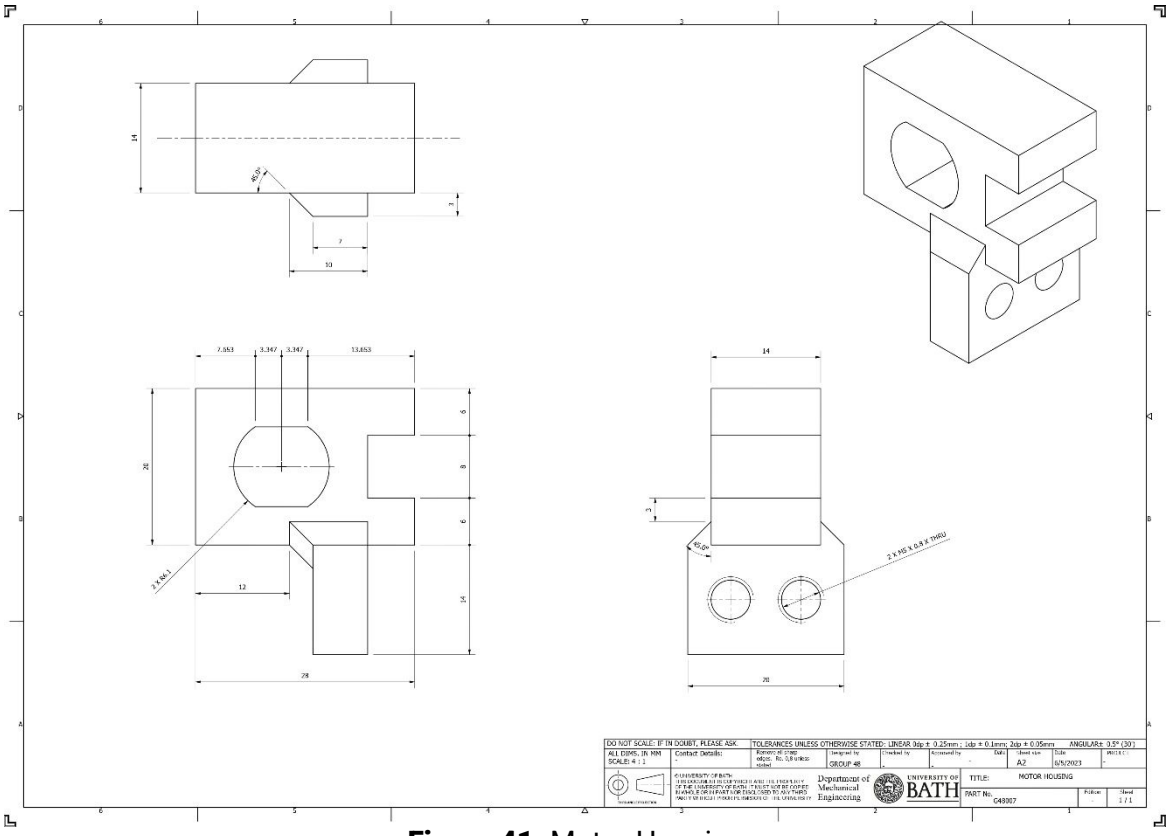


Figure 41. Motor Housing

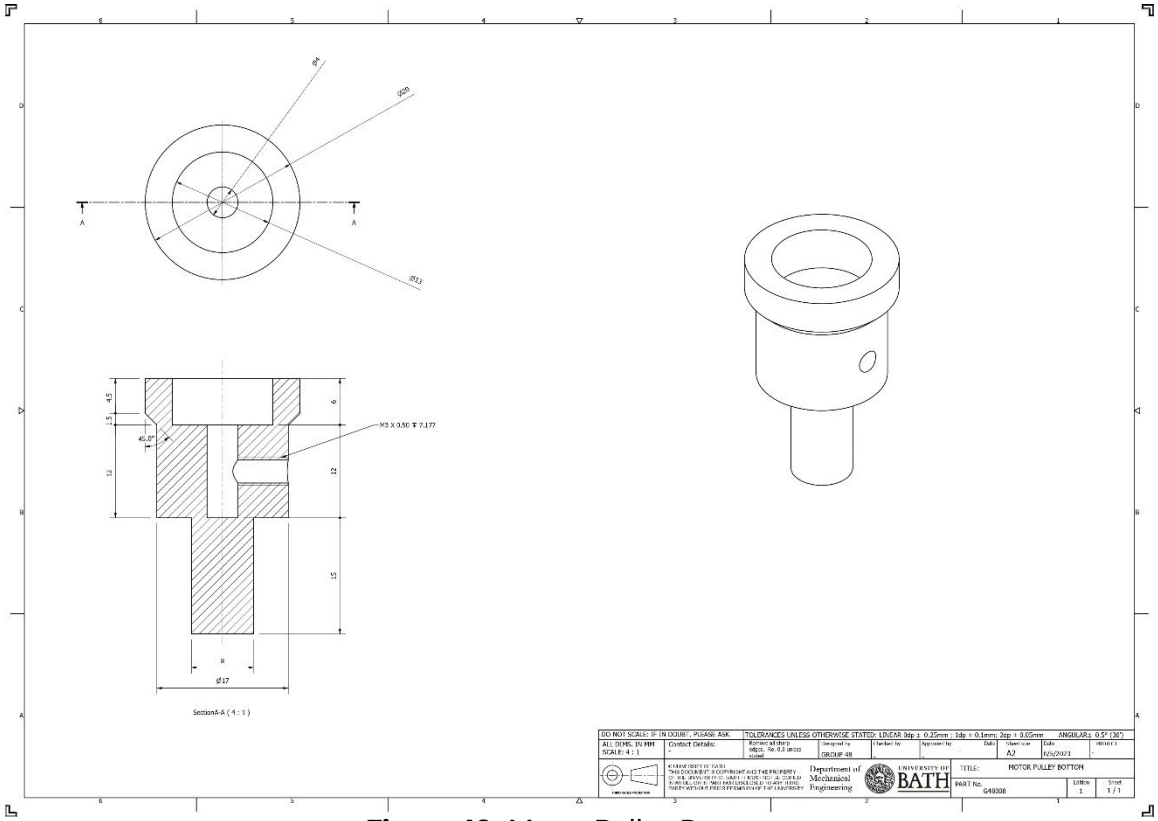


Figure 42. Motor Pulley Bottom

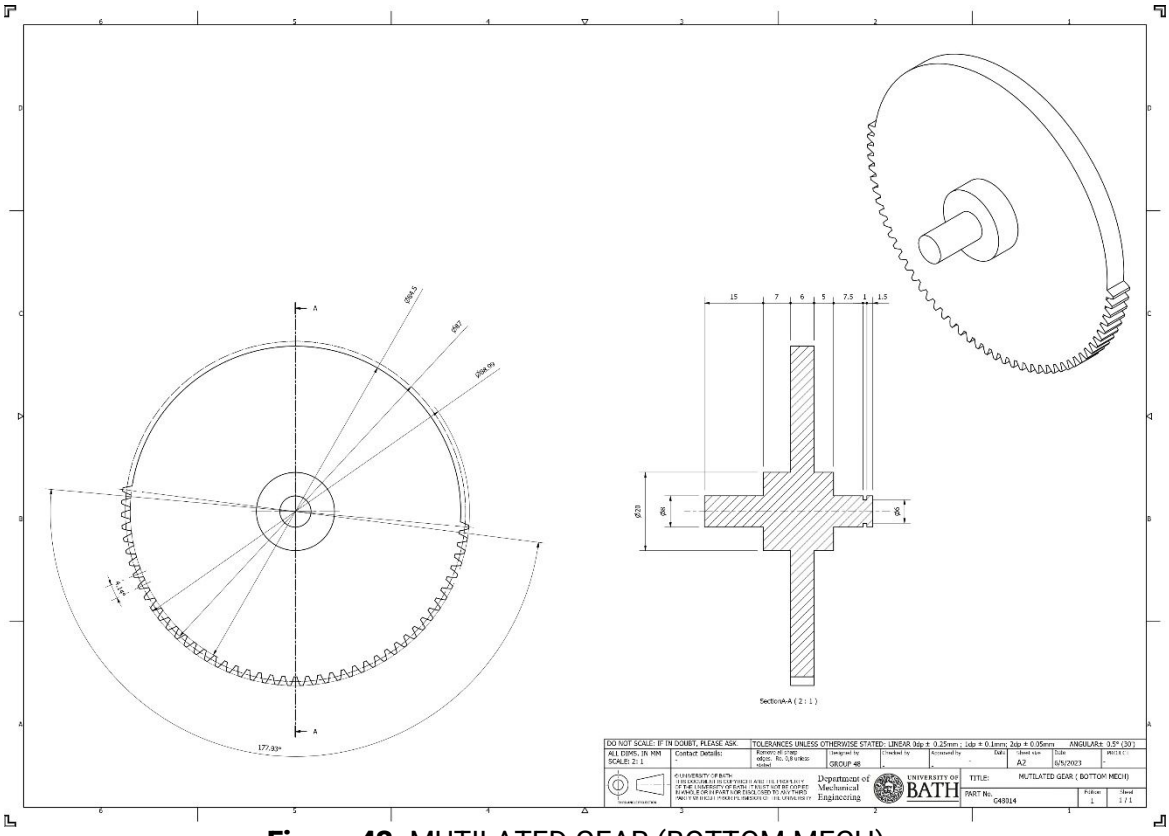


Figure 43. MUTILATED GEAR (BOTTOM MECH)

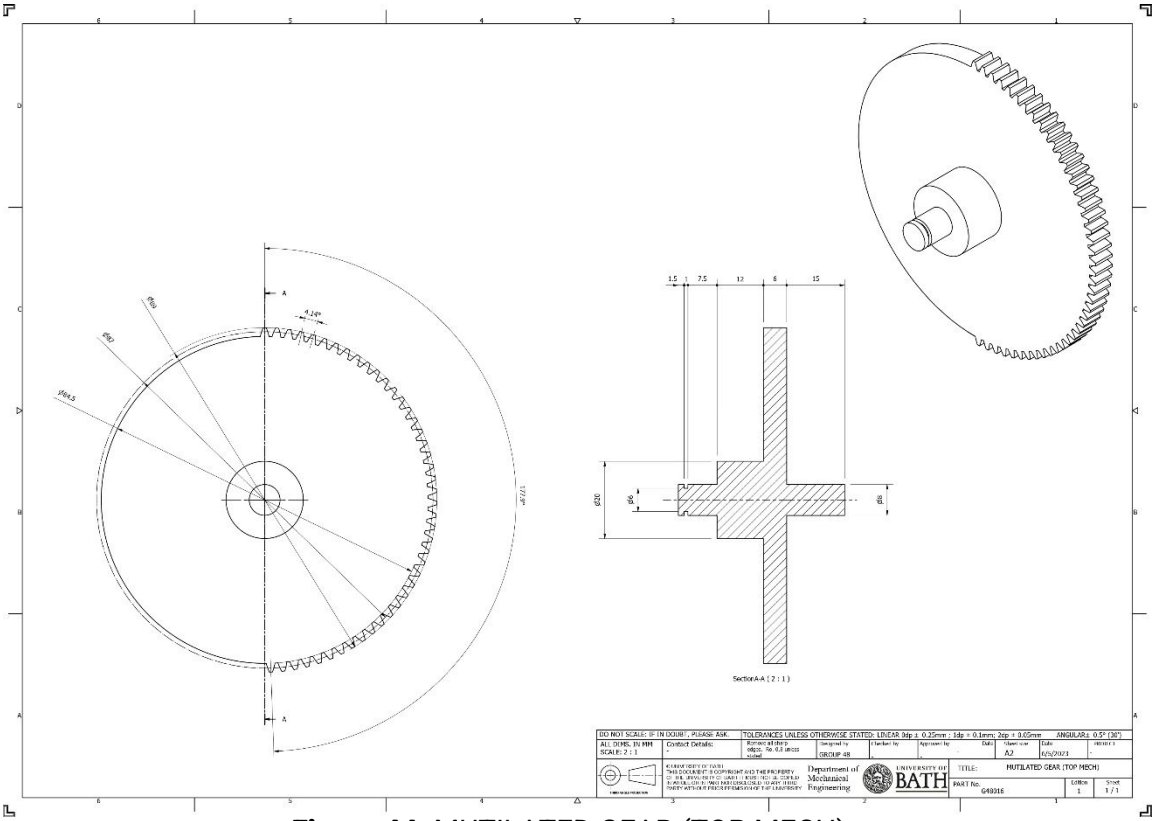


Figure 44. MUTILATED GEAR (TOP MECH)

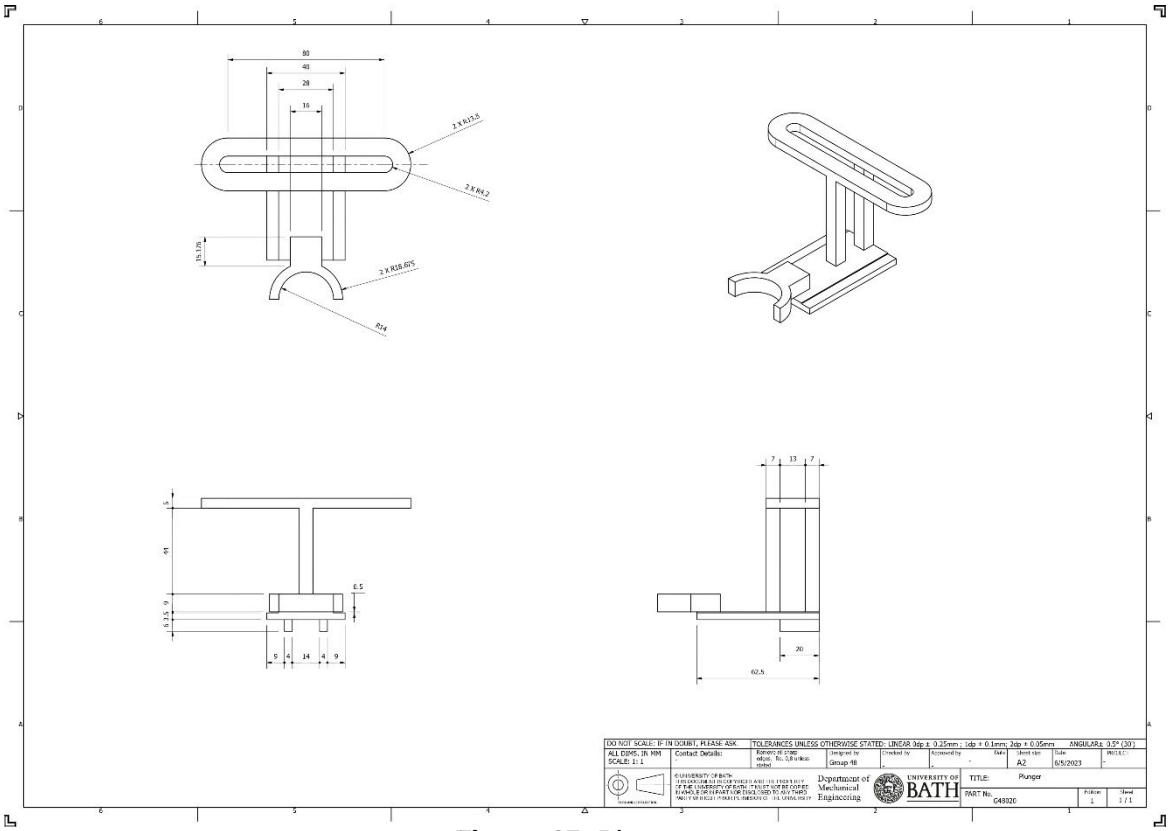


Figure 45. Plunger

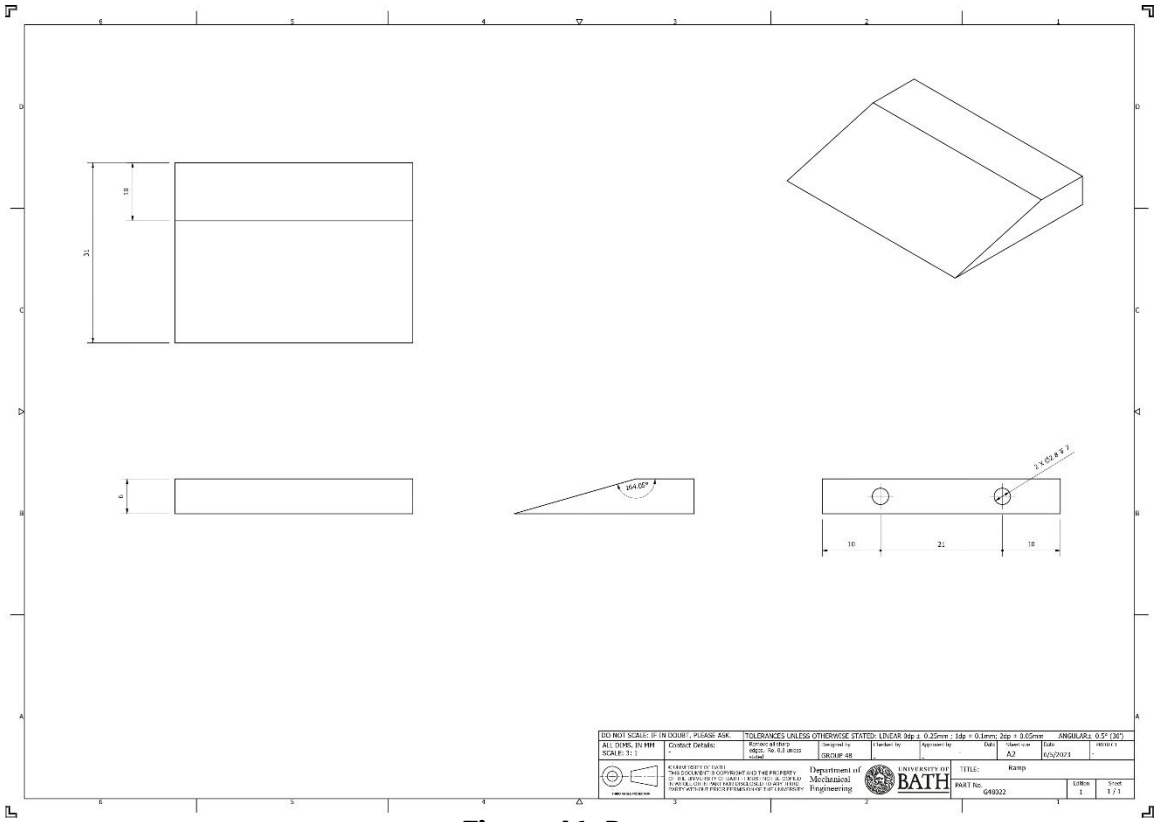


Figure 46. Ramp

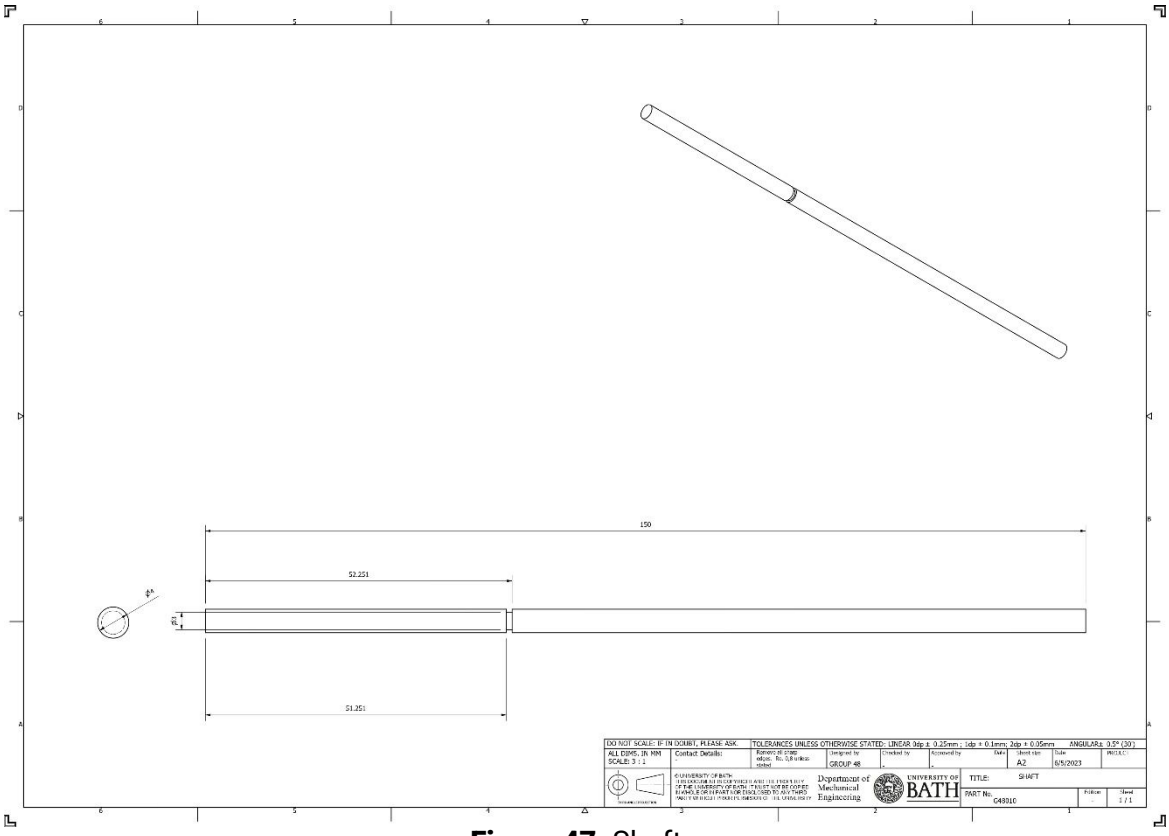


Figure 47. Shaft

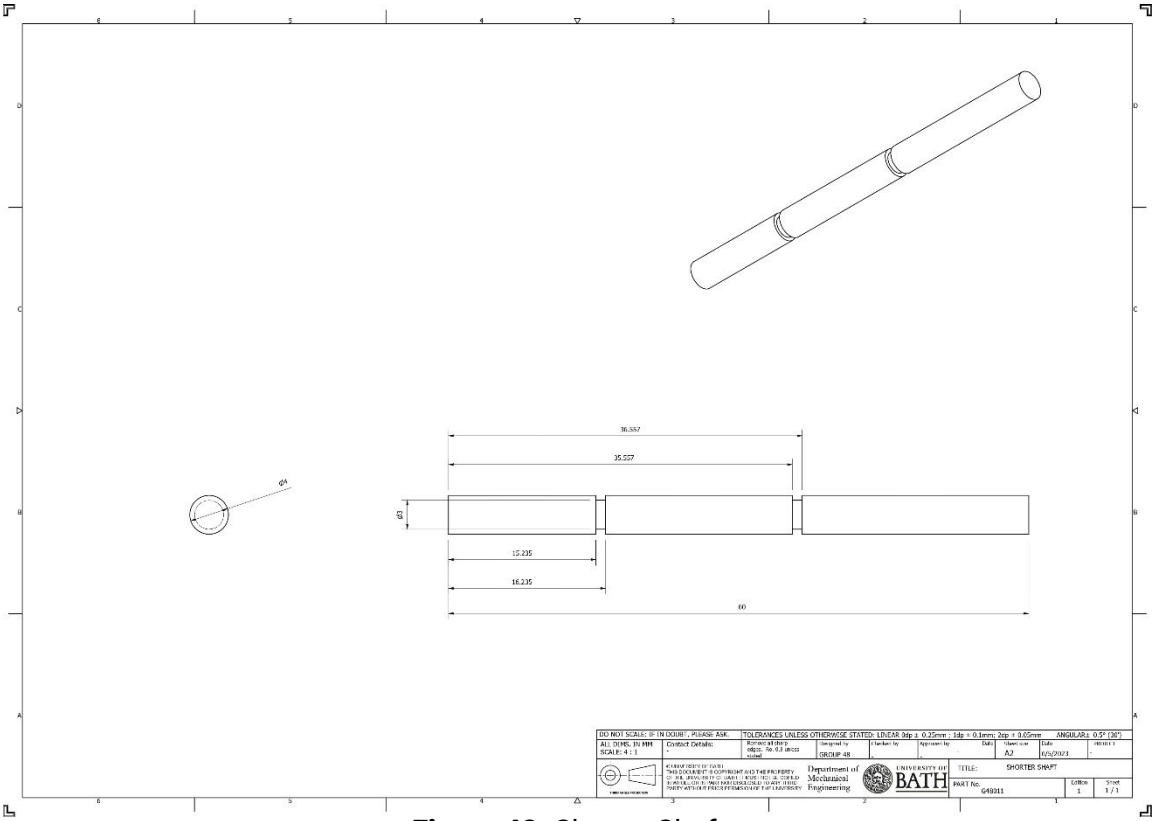


Figure 48. Shorter Shaft

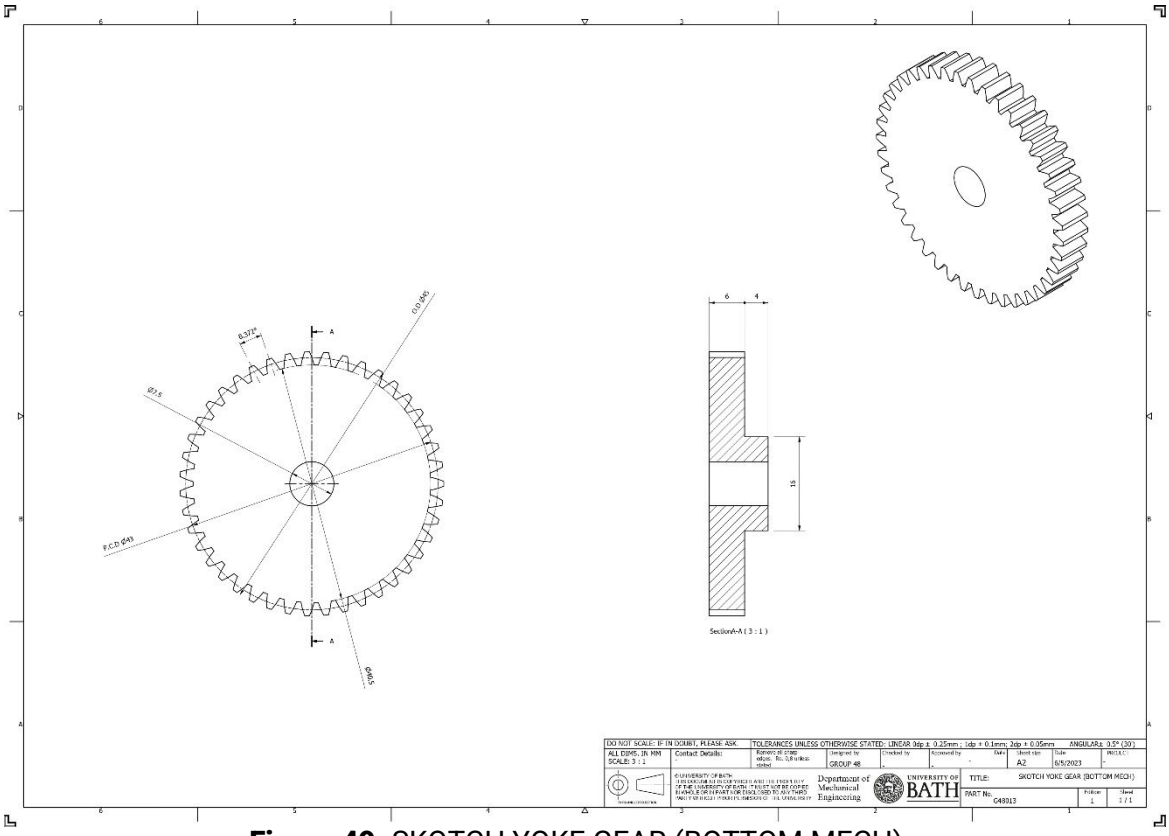


Figure 49. SKOTCH YOKE GEAR (BOTTOM MECH)

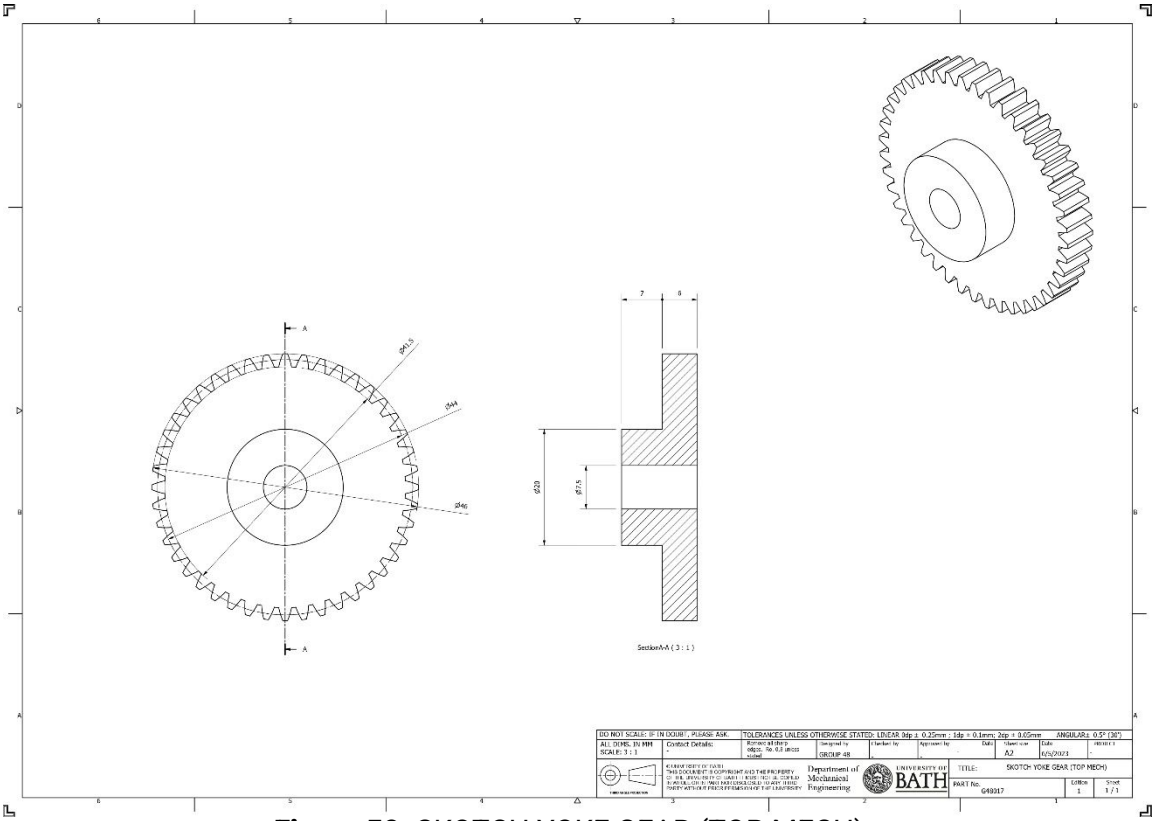


Figure 50. SKOTCH YOKE GEAR (TOP MECH)

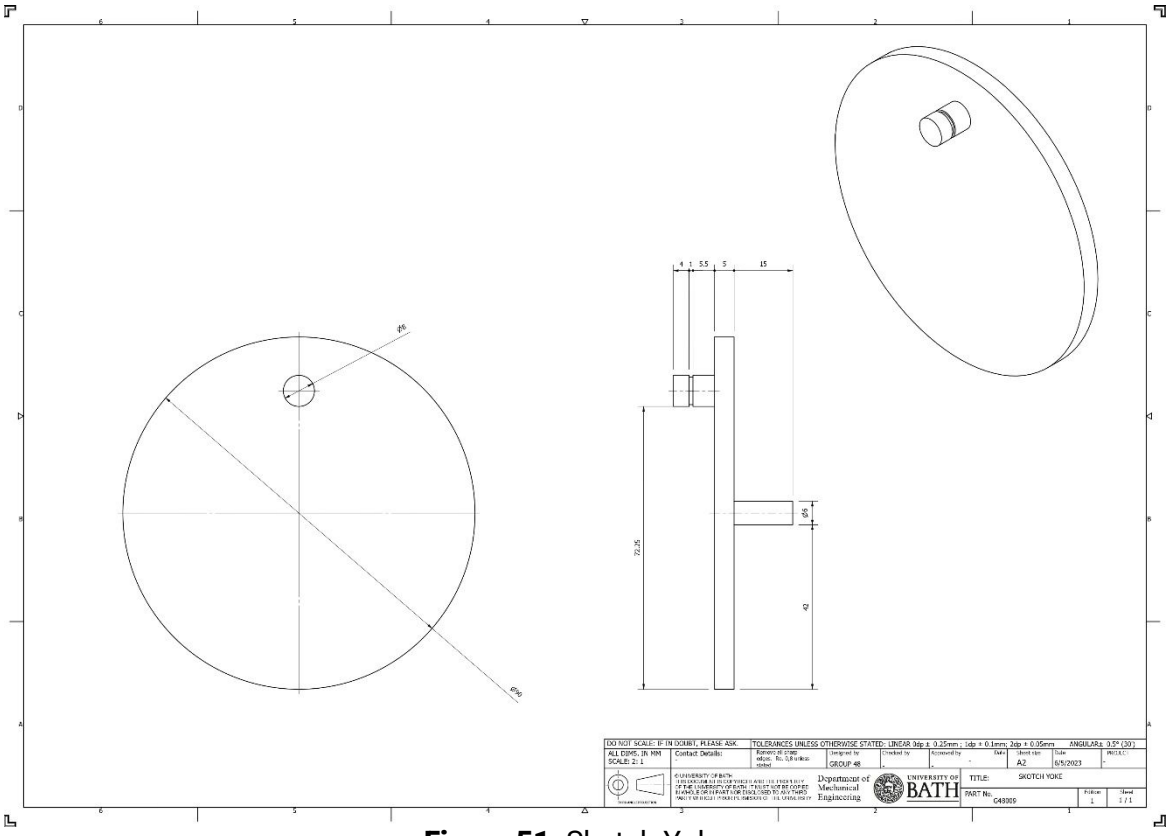


Figure 51. Skotch Yoke

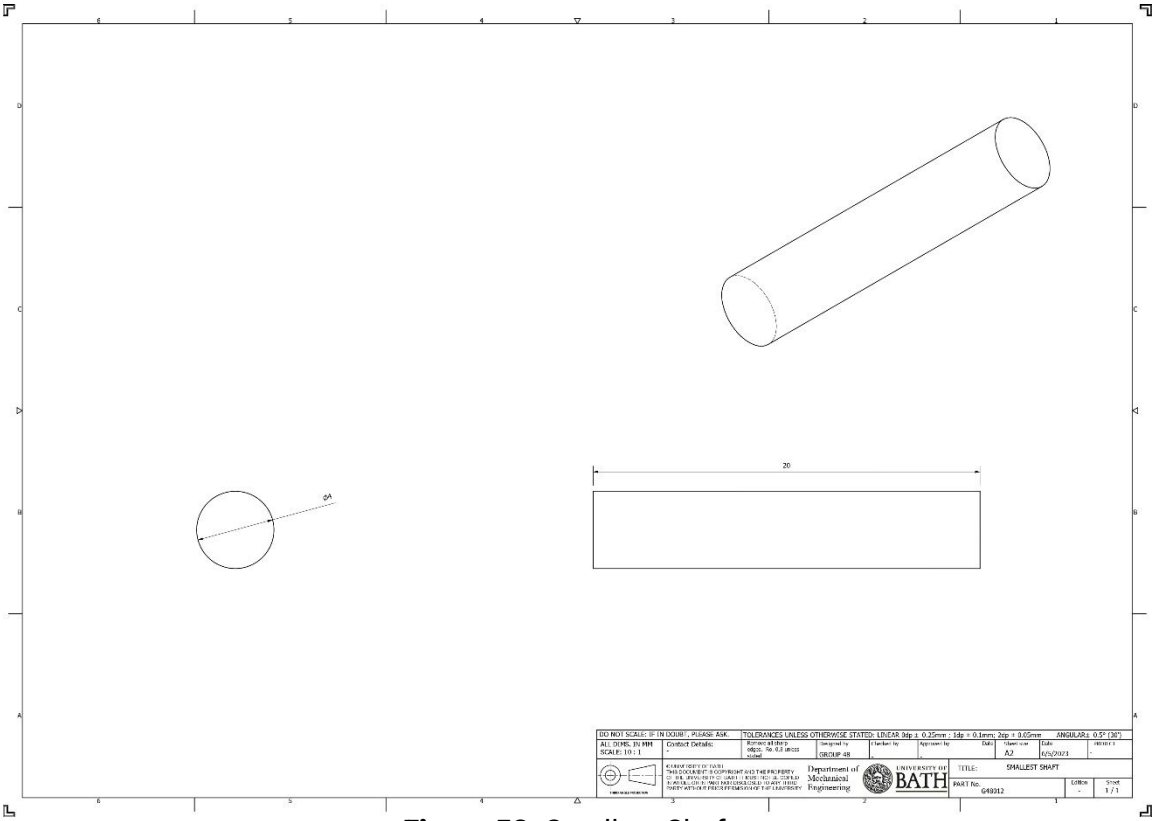


Figure 52. Smallest Shaft

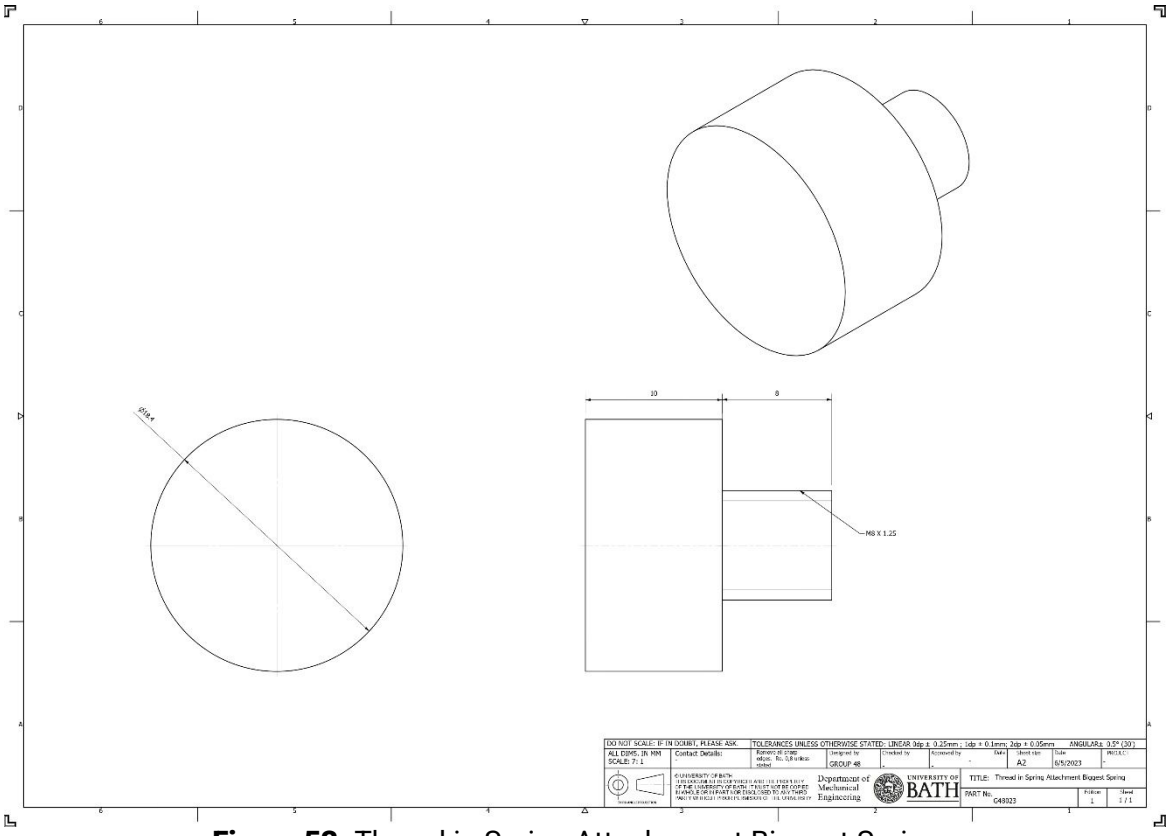


Figure 53. Thread in Spring Attachement Biggest Spring

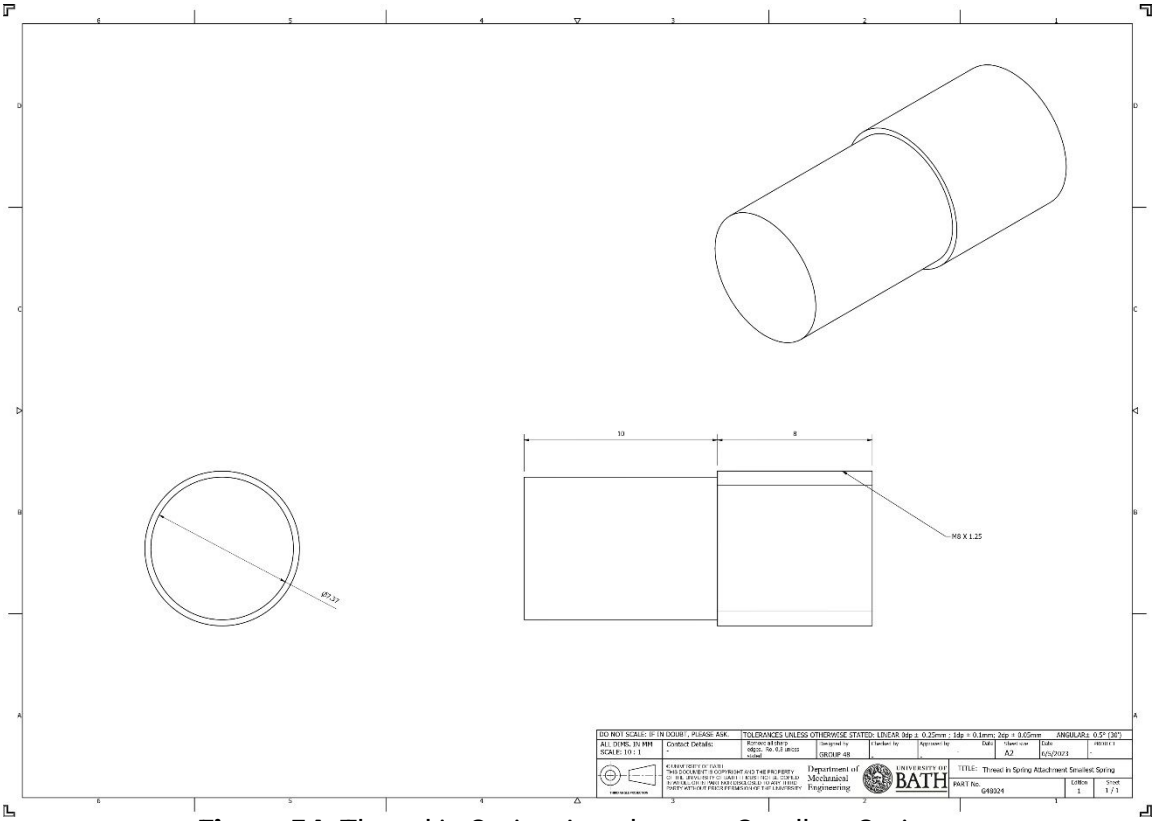


Figure 54. Thread in Spring Attachement Smallest Spring

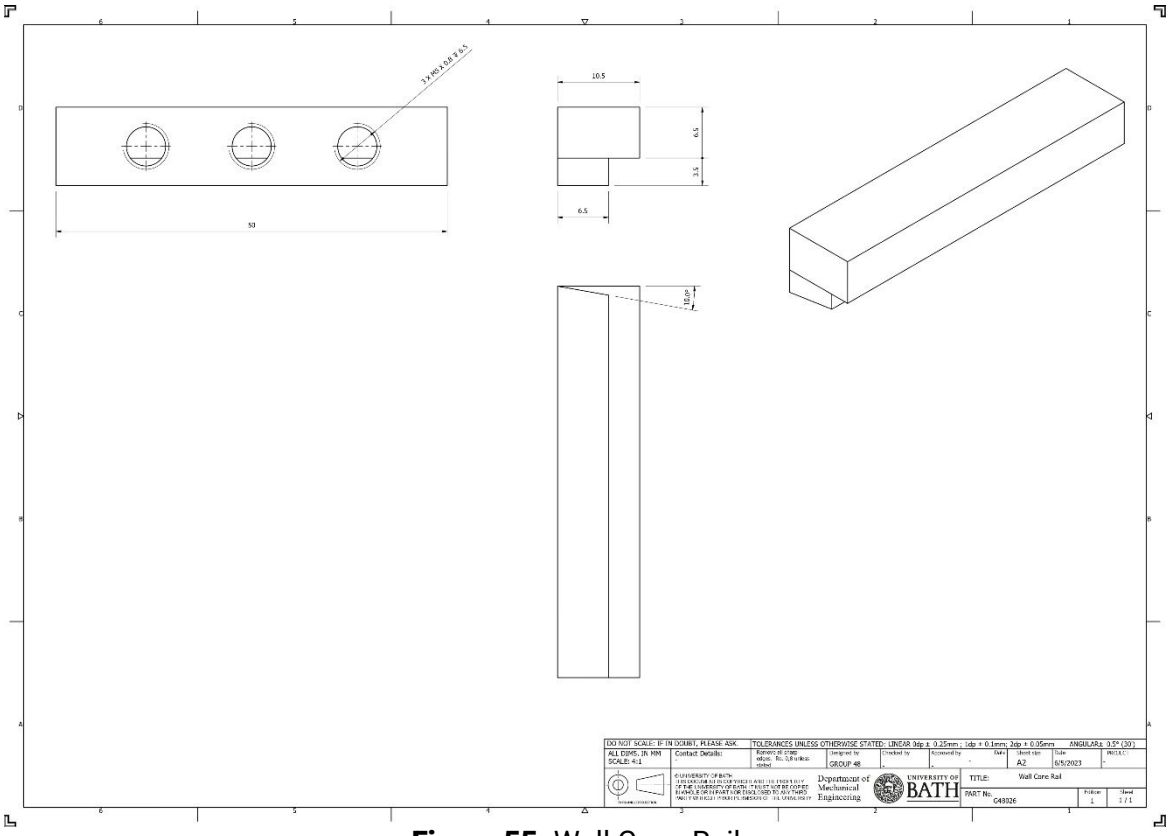


Figure 55. Wall Cone Rail

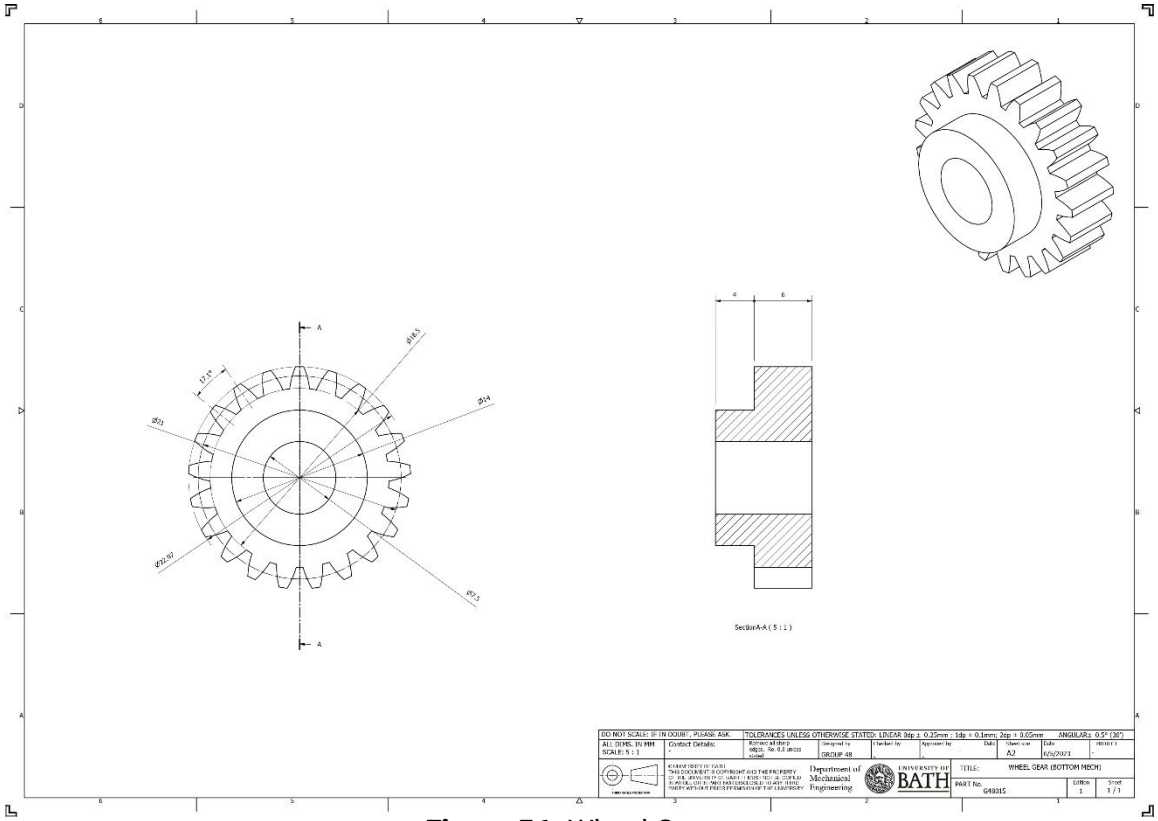


Figure 56. Wheel Gear

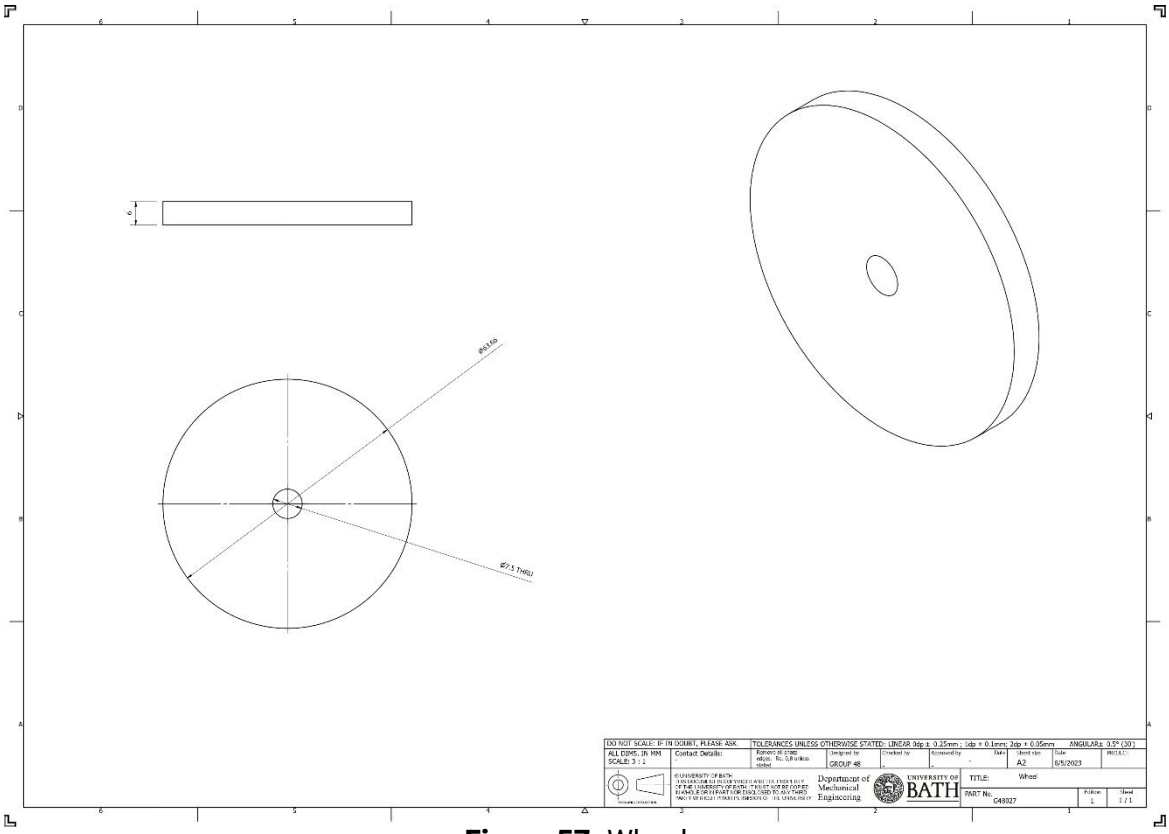


Figure 57. Wheel

Appendix D: Contact Details

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